# Methods in Human Geography

## Quantitative Methods: Foundations




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# This week

## Part I

- Module overview.
- Introduction to Quantitative Research methods.

## Part II

- Describing data.
- R for data analysis.

# Module overview

# Module overview



# Teaching format

- Lectures: Concepts and the bigger picture (Tuesday).
- Tutorials: Hands-on working with quantitative data (Wednesday, Thursday).
- Homework: Reading material, short assignments.

# Practicalities

- Lecture slides will be available online before each lecture.
- During the computer tutorials you are expected to work independently through the material for that week – main opportunity to ask questions and get assistance.
- Output of the homework tasks at the end of each of the computer tutorials is to be included in the appendix of your assignment.

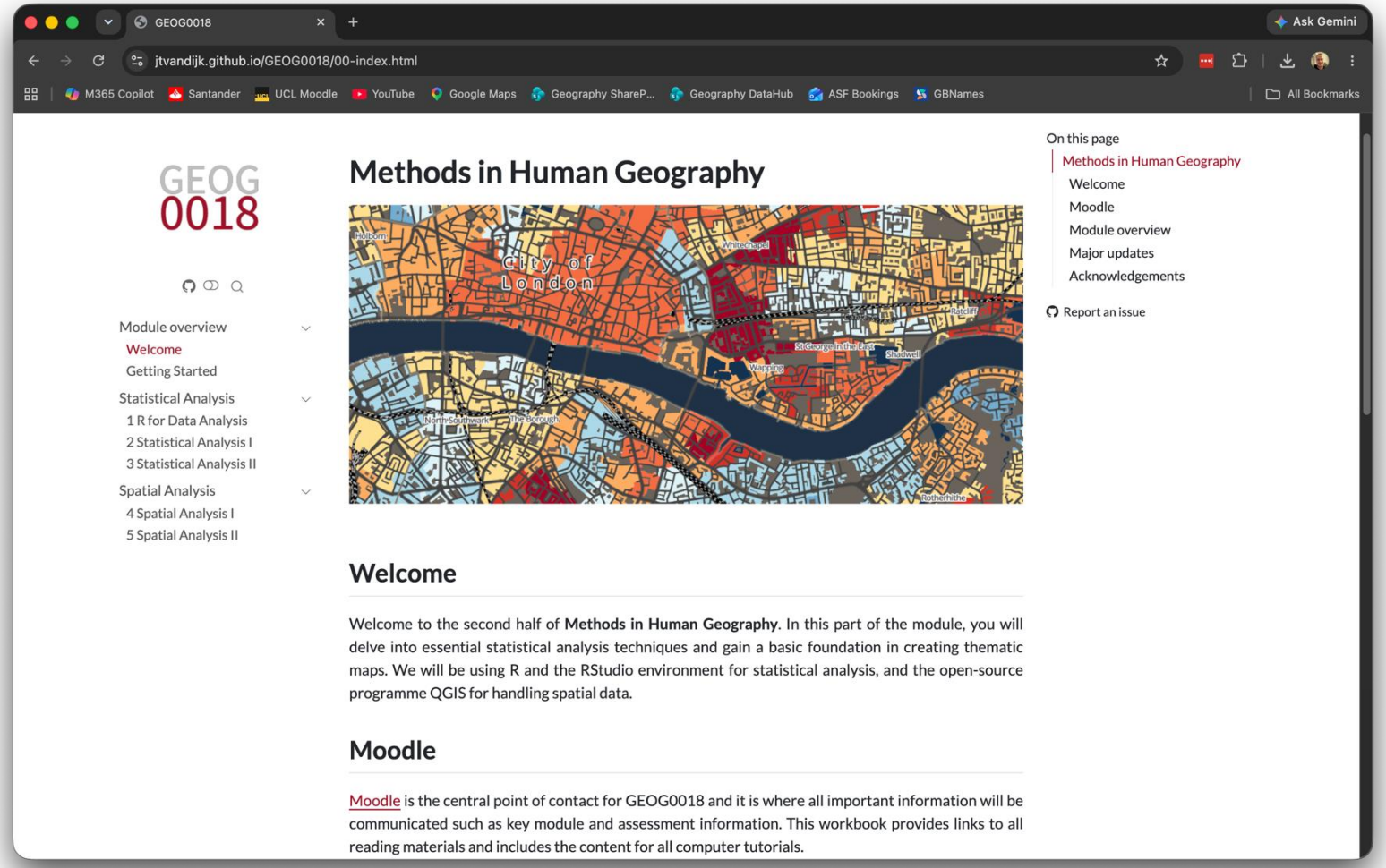
# Assessment

To explore a geographic theme (e.g. gentrification, identity, migration) in the context of London, submitted as a 1,500-word report (50% of module grade):

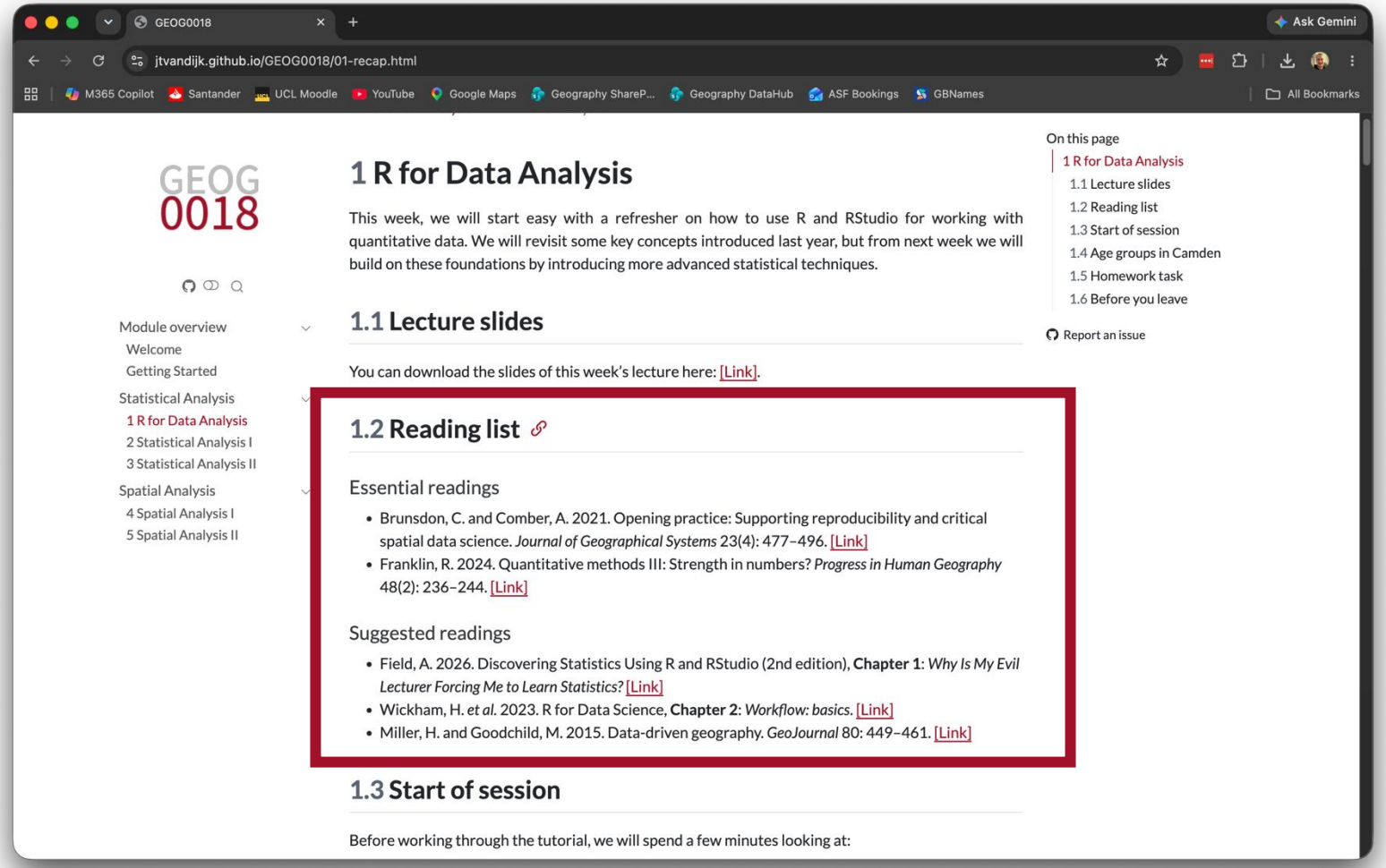
1. Outline a research question and hypothesis (200-250 words).
2. Obtain data and identify appropriate methods to use (200-250 words).
3. Present and interpret results (700-800 words).
4. Briefly evaluate your approach (200-300 words).

Deadline 14th of January 2027 – 13h00.

# Workbook


GEOG0018
Ask Gemini
jtvandijk.github.io/GEOG0018/00-index.html
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
GEOG
0018
Module overview
Welcome
Getting Started
Statistical Analysis
1 R for Data Analysis
2 Statistical Analysis I
3 Statistical Analysis II
Spatial Analysis
4 Spatial Analysis I
5 Spatial Analysis II
Methods in Human Geography
City of London
Whitechapel
Wapping
Shadwell
Rotherhithe
Welcome
Welcome to the second half of Methods in Human Geography. In this part of the module, you will delve into essential statistical analysis techniques and gain a basic foundation in creating thematic maps. We will be using R and the RStudio environment for statistical analysis, and the open-source programme QGIS for handling spatial data.
Moodle
Moodle is the central point of contact for GEOG0018 and it is where all important information will be communicated such as key module and assessment information. This workbook provides links to all reading materials and includes the content for all computer tutorials.
On this page
Methods in Human Geography
Welcome
Moodle
Module overview
Major updates
Acknowledgements
Report an issue

# Workbook


GEOG0018
Ask Gemini
jtvandijk.github.io/GEOG0018/01-recap.html
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
GEOG 0018
Module overview
Welcome
Getting Started
Statistical Analysis
1 R for Data Analysis
2 Statistical Analysis I
3 Statistical Analysis II
Spatial Analysis
4 Spatial Analysis I
5 Spatial Analysis II
1 R for Data Analysis
This week, we will start easy with a refresher on how to use R and RStudio for working with quantitative data. We will revisit some key concepts introduced last year, but from next week we will build on these foundations by introducing more advanced statistical techniques.
1.1 Lecture slides
You can download the slides of this week's lecture here: [Link].
1.2 Reading list
Essential readings
Brunsdon, C. and Comber, A. 2021. Opening practice: Supporting reproducibility and critical spatial data science. Journal of Geographical Systems 23(4): 477–496. [Link]
Franklin, R. 2024. Quantitative methods III: Strength in numbers? Progress in Human Geography 48(2): 236–244. [Link]
Suggested readings
Field, A. 2026. Discovering Statistics Using R and RStudio (2nd edition), Chapter 1: Why Is My Evil Lecturer Forcing Me to Learn Statistics? [Link]
Wickham, H. et al. 2023. R for Data Science, Chapter 2: Workflow: basics. [Link]
Miller, H. and Goodchild, M. 2015. Data-driven geography. GeoJournal 80: 449–461. [Link]
1.3 Start of session
Before working through the tutorial, we will spend a few minutes looking at:
On this page
1 R for Data Analysis
1.1 Lecture slides
1.2 Reading list
1.3 Start of session
1.4 Age groups in Camden
1.5 Homework task
1.6 Before you leave
Report an issue

# Communication

- All important information will be communicated through Moodle.
- Academic Support and Feedback (ASF) Office Hours:
  - Mondays 14h00–16h00 [North–West Wing Room 118].
  - Thursdays 13h00–14h00 [North–West Wing Room 118].
  - Slots can be booked through Microsoft Bookings: [Link]

# Troubleshooting and support

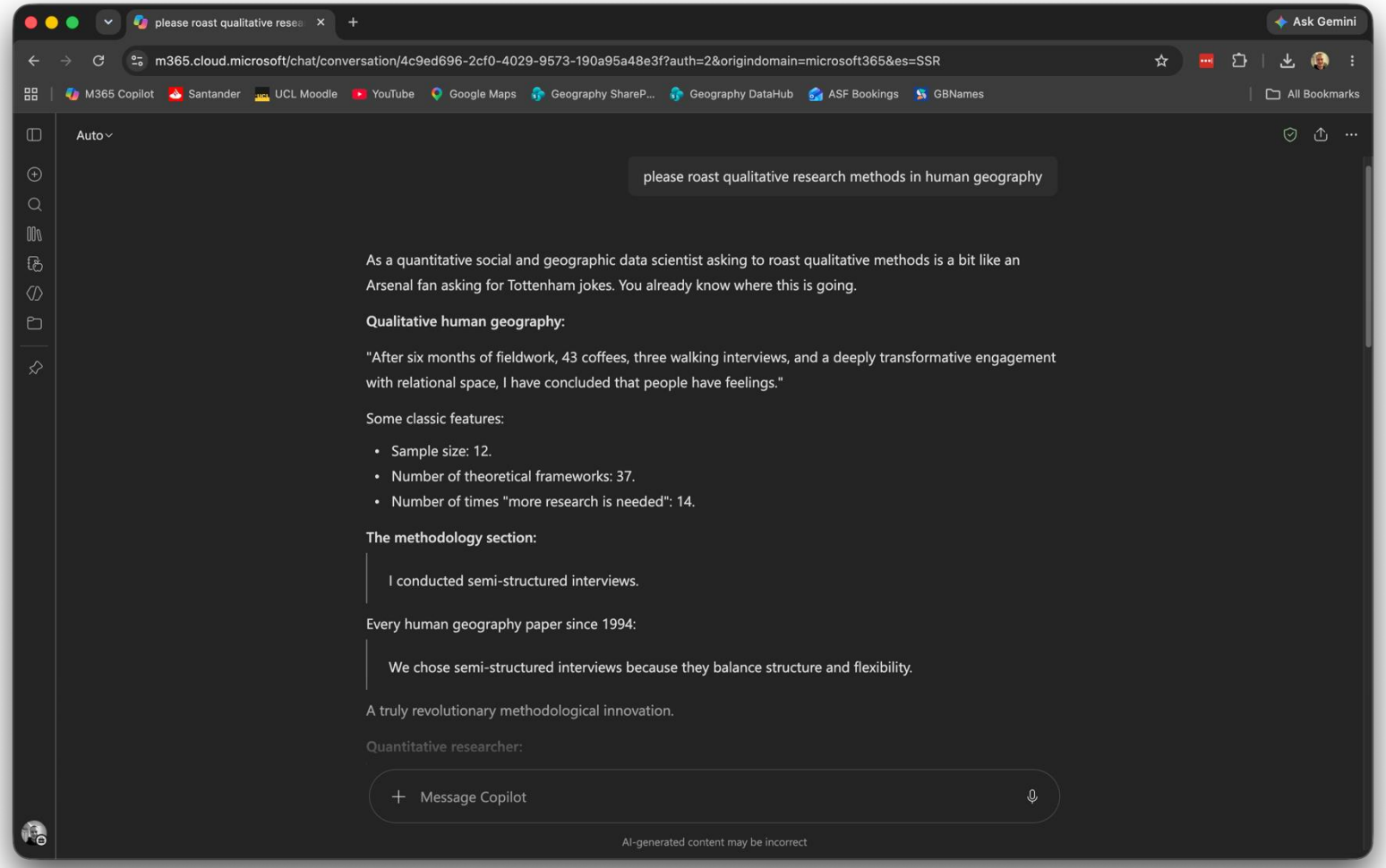
- Come prepared to the weekly computer tutorials.
- Drop by the Departmental Coding Therapy sessions.
  - Wednesdays 15h00–16h00 [North–West Wing Room 304].
- Make use of ASF hours.

# Mentimeter

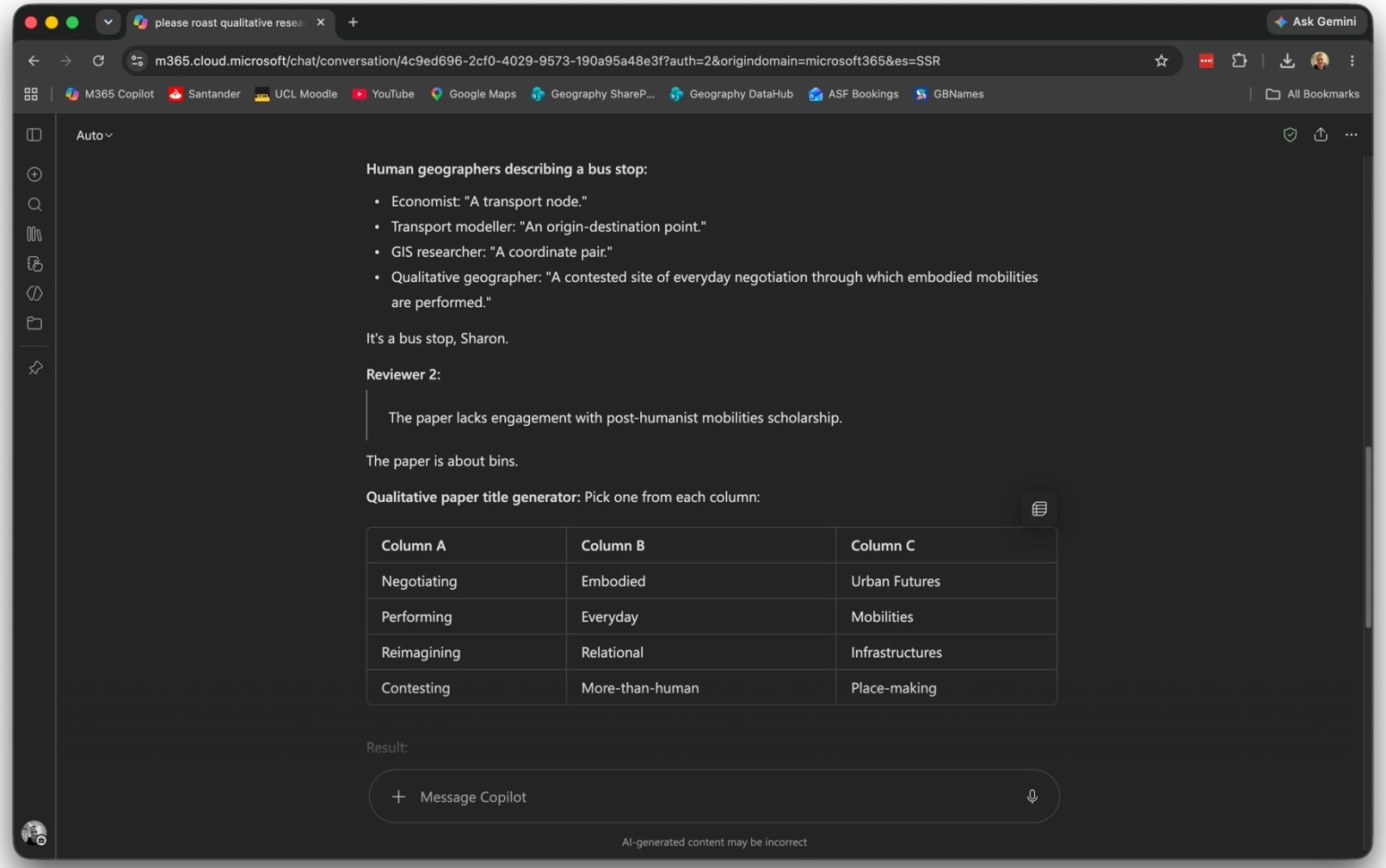
- Go to www.menti.com.
- Use code: XXXX XXXX

# Quantitative Research Methods

# Qualitative methods



# Qualitative methods



# Qualitative methods: the stylised ‘facts’

- Exploratory and descriptive in nature?
- Seeks to understand underlying reasons, opinions, and motivations?
- Depth over breadth?
- Crafting a context?
- Subjectivity and positionality?
- Non-numerical data.

# Quantitative methods: the stylised ‘facts’

- Generalising results to a broader population?
- Representation?
- Seeks to describe and understand patterns?
- Breadth (extensive) over depth (intensive)?
- Testing hypotheses and looking for relationships?
- Objective?
- Numerical data.

# Philosophical foundations

| Nomothetic geography | Idiographic geography |
| --- | --- |
| Uncover generalisable laws that explain patterns and processes. | Describe difference and particularity. |
| Focus on the population, either directly or by representative sampling. | Focus on the specific and context dependence. |
| Geographer as a detached, rational observer who uses scientific methods to ‘find and explain’. | Openness to embedded research methods, subjectivity, positionality, recognition of social constructions. |

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Where does quantitative research fit?

# A very brief history

1950s–70s: The Quantitative Revolution

- Response to the perceived lack of scientific rigour in geography.
- Sought to make geography more scientific, objective, and systematic by adopting methods from natural sciences (e.g. regression analysis, gravity models).
- Focus on discovering spatial patterns, laws, and generalisable theories.
- Schaefer (1953) rejected regional geography and called for laws. Methods spread from a few US universities (Iowa, Washington, Wisconsin).
- The philosophy came later: Harvey (1969), *Explanation in Geography*.

# A very brief history

1980s–90s: The Cultural Turn

- Response to the limitations of the Quantitative Revolution.
- Emphasis on the importance of culture, meaning, and human subjectivity in understanding socio-spatial phenomena.
- Focus on meaning and representation, attention to power and social justice (e.g. perspectives from critical theory, feminism, postcolonialism).
- Humanistic geographers criticised spatial science for dehumanising people. Radical geographers (Harvey, Bunge) turned to poverty, war and civil rights.

# A very brief history

2000s–Present: Data-driven Geography?

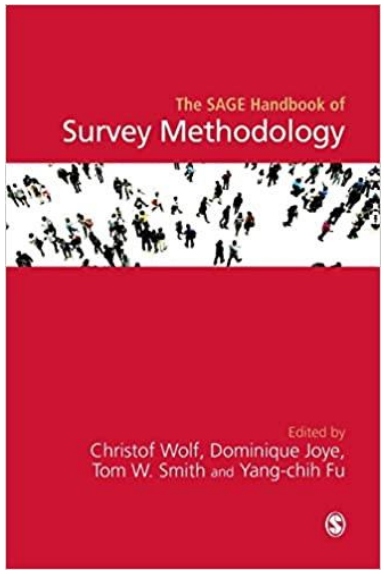
- Proliferation of data, vastly increased computational power, new methods.
- Greater sensitivity to diversity and difference.
- Geographic Data Science? Quantitative human geography? GeoAI? Urban analytics? Geocomputation? Urban data science?
- See Miller and Goodchild (2015), Franklin (2024).

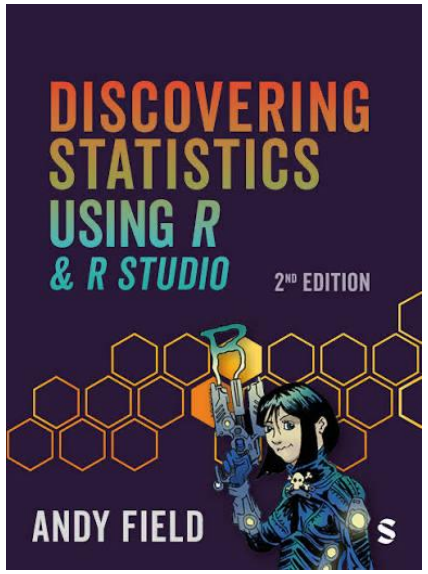
# A very brief history

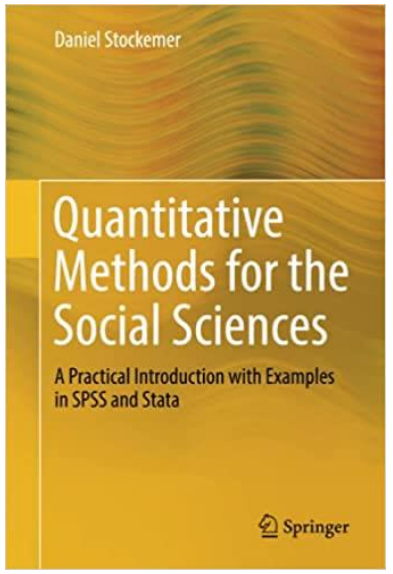








# What are data?

Kitchin 2014:

“Data are commonly understood to be the raw material produced by abstracting the world into categories, measures and other representational forms – numbers, characters, symbols, images, sounds, electromagnetic waves, bits – that constitute the building blocks from which information and knowledge are created.” (p.2)

Key point: data are produced, so we need to think carefully about this production process and its implications (for knowledge, politics, practice).

# Quantitative data

- Quantitative: Numeric, properties of phenomena.
  Measurement scale (nominal, ordinal, interval, ratio).
- Qualitative: Non-numeric.

# Quantitative data

- Structured: Data that is easily organised, defined data model.
- Semi-structured: Loosely structured data, often hierarchical (XML).
- Unstructured: Data without an identifiable structure (not per se qualitative).

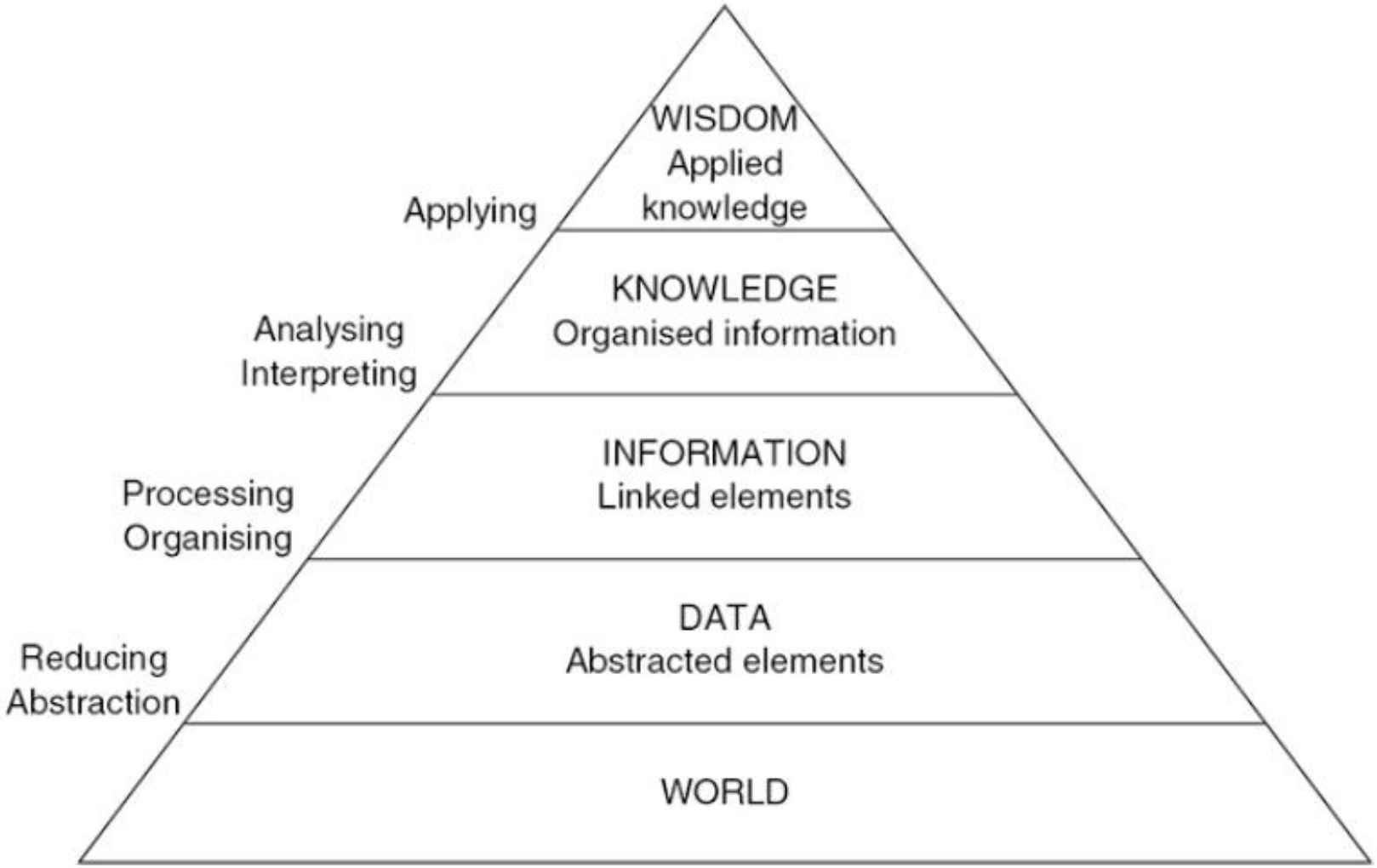
# Quantitative data

- Captured: Data that is measured with some instrument (e.g. survey).
- Exhausted: Data as a by-product (e.g. transaction data).
- Derived: Data derived from other data (e.g. model output).

# Quantitative data

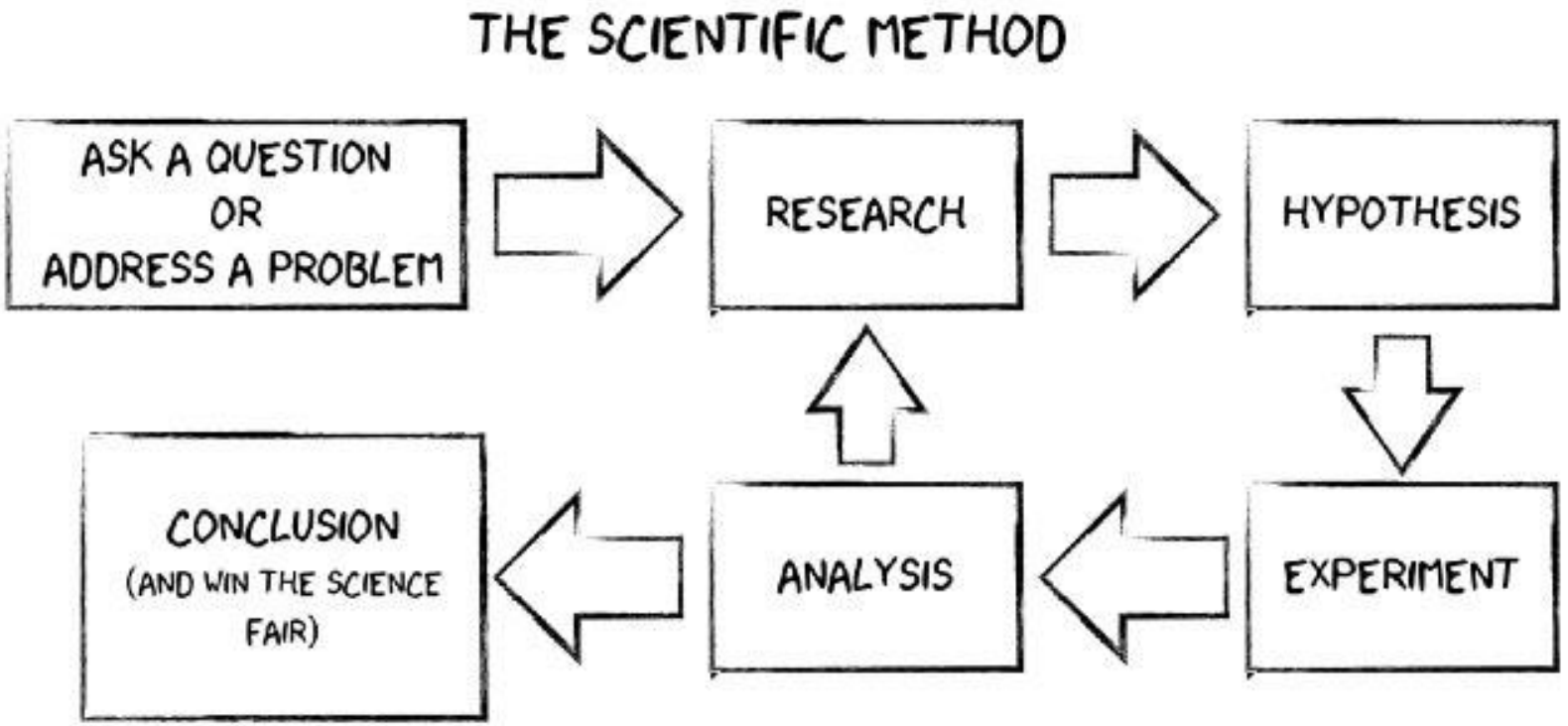
- Primary: Data captured by a researcher.
- Secondary: ‘Analysis-ready’ dataset, pre-processed, cleaned.
- Tertiary: ‘Amplified data’ (e.g. geodemographics).

# Quantitative data



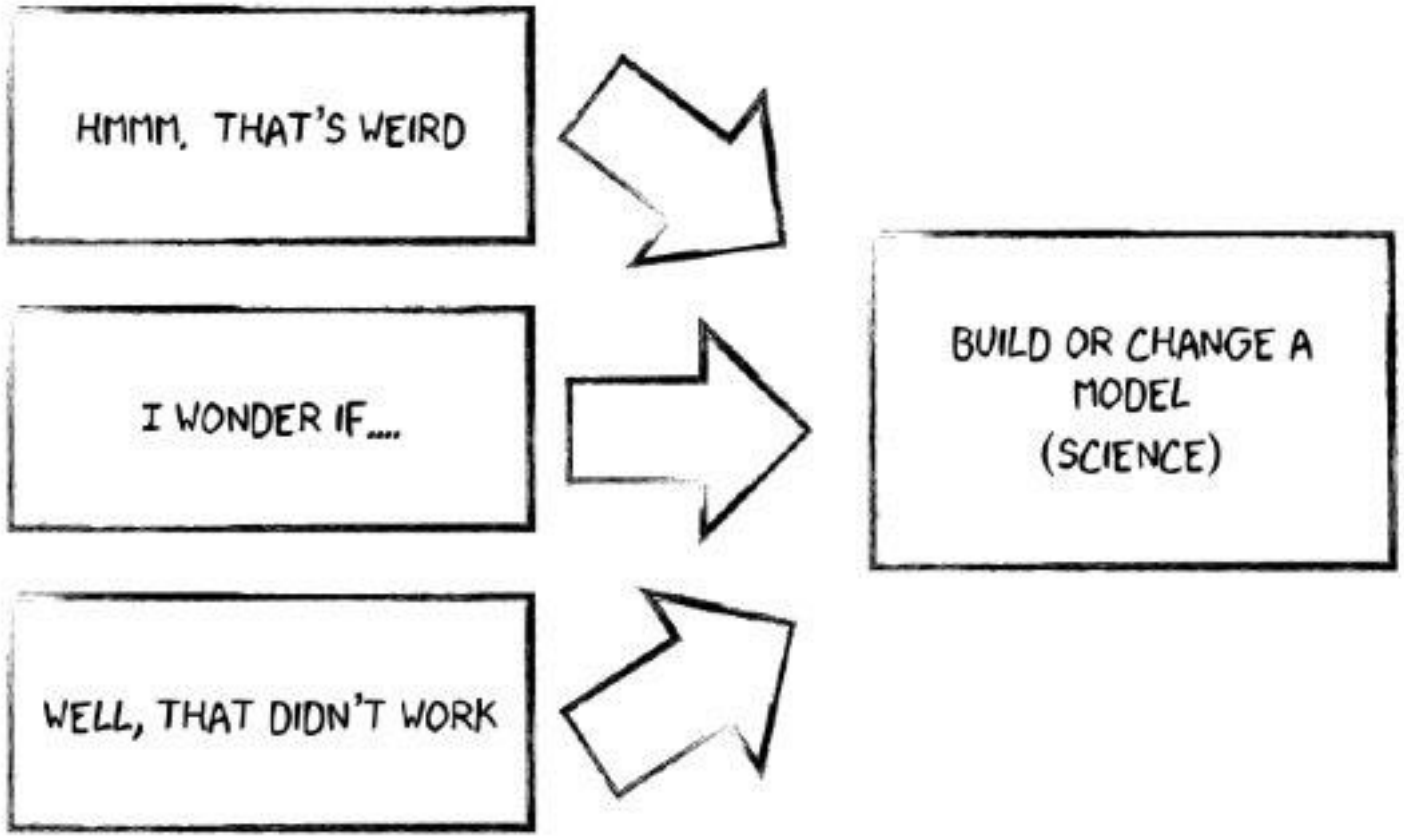

Kitchin. 2014. Knowledge pyramid adapted from Adler (1986) and McCandless (2010)

# Research questions




Allain. 2013. What's wrong with the scientific method? *Wired* 01/04/2013.

# Starting from the middle




Allain. 2013. What's wrong with the scientific method? *Wired* 01/04/2013.

# Research questions

Research questions should:

- Contribute to theory, improve methods, or provide new evidence.
- Build on existing knowledge.
- Be concisely and precisely stated.
- Be feasible given the time, resources, and data available.

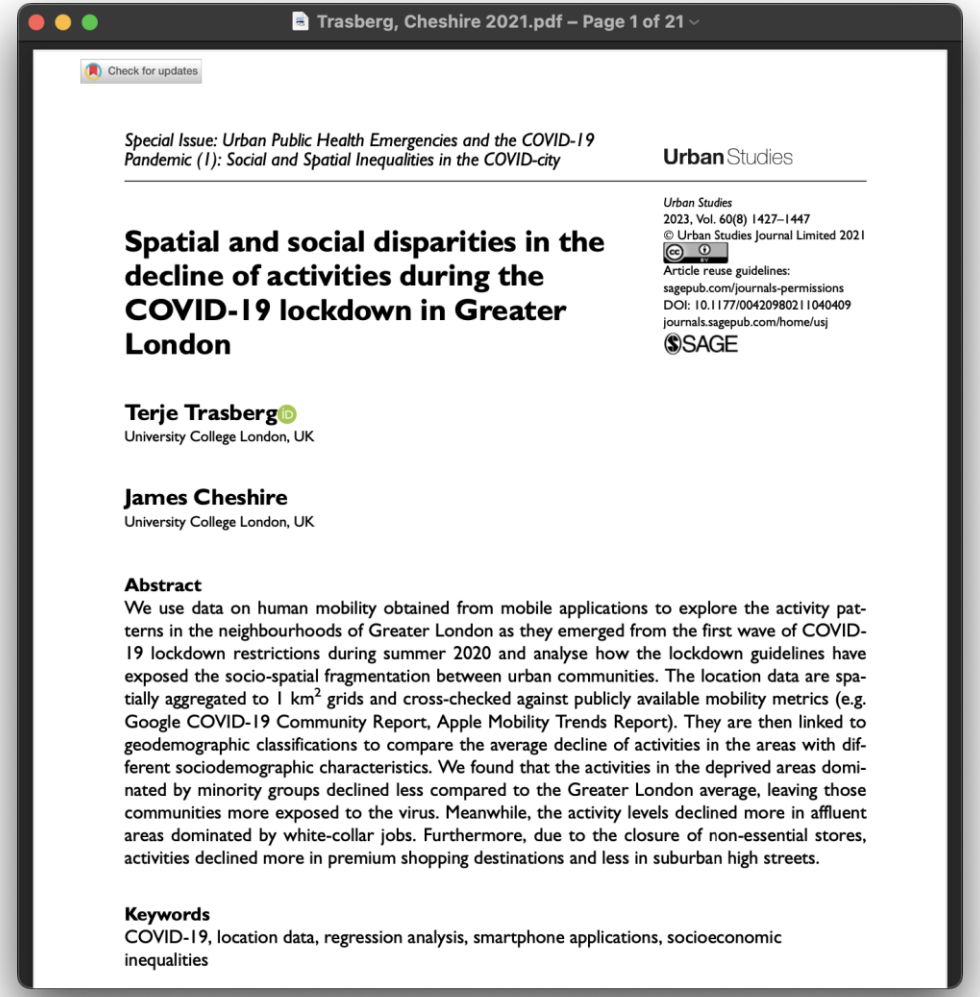
‘No one has done this before’ is necessary but not sufficient. So what?

# Research questions

Research questions can:

- Aim to describe a phenomenon.
- Compare different groups or variables.
- Seek to explain causes or effects.
- Investigate new or under-researched areas.
- Provide a new way of thinking about or looking at a problem.
- Assess the effectiveness of a policy or intervention.

# Research examples



Trasberg, Cheshire 2021.pdf – Page 1 of 21

Check for updates

*Special Issue: Urban Public Health Emergencies and the COVID-19 Pandemic (1): Social and Spatial Inequalities in the COVID-city*

Urban Studies

*Urban Studies*
2023, Vol. 60(8) 1427–1447
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SAGE

## Spatial and social disparities in the decline of activities during the COVID-19 lockdown in Greater London

**Terje Trasberg**
University College London, UK

**James Cheshire**
University College London, UK

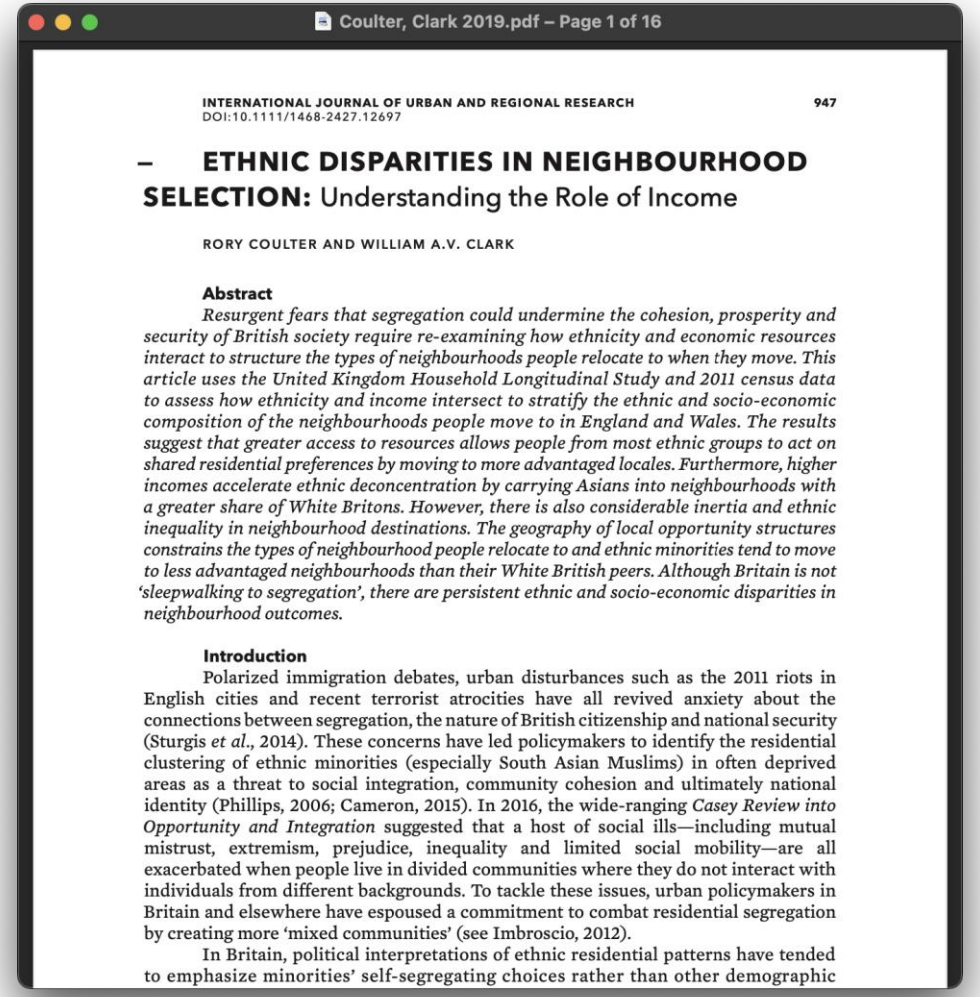
**Abstract**

We use data on human mobility obtained from mobile applications to explore the activity patterns in the neighbourhoods of Greater London as they emerged from the first wave of COVID-19 lockdown restrictions during summer 2020 and analyse how the lockdown guidelines have exposed the socio-spatial fragmentation between urban communities. The location data are spatially aggregated to 1 $km^2$ grids and cross-checked against publicly available mobility metrics (e.g. Google COVID-19 Community Report, Apple Mobility Trends Report). They are then linked to geodemographic classifications to compare the average decline of activities in the areas with different sociodemographic characteristics. We found that the activities in the deprived areas dominated by minority groups declined less compared to the Greater London average, leaving those communities more exposed to the virus. Meanwhile, the activity levels declined more in affluent areas dominated by white-collar jobs. Furthermore, due to the closure of non-essential stores, activities declined more in premium shopping destinations and less in suburban high streets.

**Keywords**

COVID-19, location data, regression analysis, smartphone applications, socioeconomic inequalities

# Research examples




INTERNATIONAL JOURNAL OF URBAN AND REGIONAL RESEARCH
DOI:10.1111/1468-2427.12697


## ETHNIC DISPARITIES IN NEIGHBOURHOOD SELECTION: Understanding the Role of Income

RORY COULTER AND WILLIAM A.V. CLARK

**Abstract**

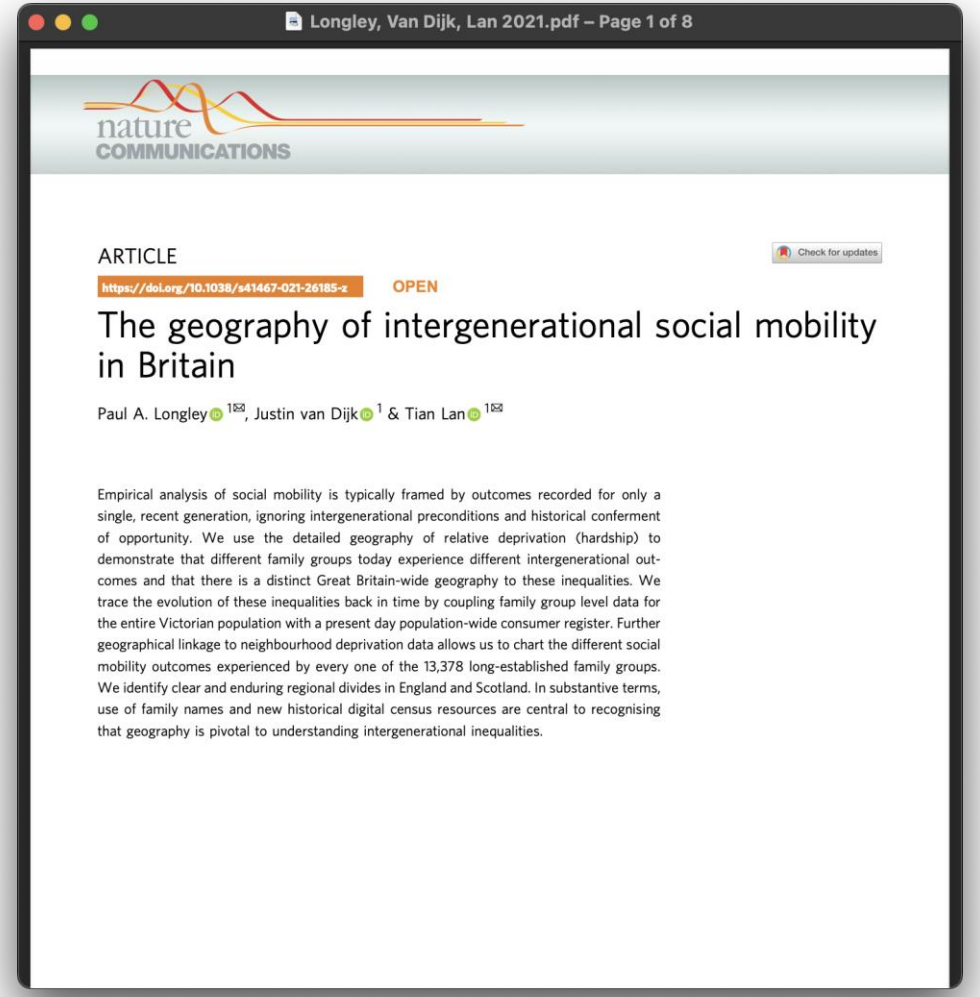
*Resurgent fears that segregation could undermine the cohesion, prosperity and security of British society require re-examining how ethnicity and economic resources interact to structure the types of neighbourhoods people relocate to when they move. This article uses the United Kingdom Household Longitudinal Study and 2011 census data to assess how ethnicity and income intersect to stratify the ethnic and socio-economic composition of the neighbourhoods people move to in England and Wales. The results suggest that greater access to resources allows people from most ethnic groups to act on shared residential preferences by moving to more advantaged locales. Furthermore, higher incomes accelerate ethnic deconcentration by carrying Asians into neighbourhoods with a greater share of White Britons. However, there is also considerable inertia and ethnic inequality in neighbourhood destinations. The geography of local opportunity structures constrains the types of neighbourhood people relocate to and ethnic minorities tend to move to less advantaged neighbourhoods than their White British peers. Although Britain is not 'sleepwalking to segregation', there are persistent ethnic and socio-economic disparities in neighbourhood outcomes.*

### Introduction

Polarized immigration debates, urban disturbances such as the 2011 riots in English cities and recent terrorist atrocities have all revived anxiety about the connections between segregation, the nature of British citizenship and national security (Sturgis *et al.*, 2014). These concerns have led policymakers to identify the residential clustering of ethnic minorities (especially South Asian Muslims) in often deprived areas as a threat to social integration, community cohesion and ultimately national identity (Phillips, 2006; Cameron, 2015). In 2016, the wide-ranging *Casey Review into Opportunity and Integration* suggested that a host of social ills—including mutual mistrust, extremism, prejudice, inequality and limited social mobility—are all exacerbated when people live in divided communities where they do not interact with individuals from different backgrounds. To tackle these issues, urban policymakers in Britain and elsewhere have espoused a commitment to combat residential segregation by creating more 'mixed communities' (see Imbroscio, 2012).

In Britain, political interpretations of ethnic residential patterns have tended to emphasize minorities' self-segregating choices rather than other demographic

# Research examples



# Break

# Describing data

# Variables

- A variable is a characteristic that can take different values, for example height, temperature or travel mode.
- A variable is measured for each unit of analysis: a person, household, neighbourhood, firm, region or country.
- Variables are used to test hypotheses and answer questions.
- Dependent variable ($\mathrm{y}$): the phenomenon we want to describe, explain, or predict.
- Independent variables ($x$): the things that might cause or affect variations in $Y$.
- Variables are measured at different levels: nominal, ordinal, interval and ratio.

# Nominal variables

- A nominal variable has categories or labels with no inherent order or ranking.
- Categories are mutually exclusive.

# Nominal variables

- A nominal variable has categories or labels with no inherent order or ranking.
- Categories are mutually exclusive.
- “How do you normally travel to UCL?”

# Nominal variables

| Respondent | Answer |
|---|---|
| 1 | Bus |
| 2 | Bike |
| 3 | Tube |
| 4 | Tube |
| 5 | Tube |
| 6 | Walk |
| 7 | Walk |
| 8 | Bike |

# Ordinal variables

- An ordinal variable has categories with a clear order or ranking.
- Intervals between the categories are not necessarily equal.

# Ordinal variables

- An ordinal variable has categories with a clear order or ranking.
- Intervals between the categories are not necessarily equal.
- “How likely are you to use quantitative methods in your dissertation?”

# Ordinal variables

| Respondent | Answer* |
|---|---|
| 1 | 1 **2** 3 4 5 |
| 2 | 1 2 3 **4** 5 |
| 3 | 1 2 3 4 **5** |
| 4 | 1 2 3 4 **5** |
| 5 | 1 2 3 **4** 5 |
| 6 | 1 2 3 **4** 5 |
| 7 | 1 2 3 4 **5** |
| 8 | **1** 2 3 4 5 |

* 1 is very likely, 2 is likely, 3 is neutral, 4 is unlikely, 5 is very unlikely

# Interval variables

- An interval variable is numeric and the intervals between values are equal.
- An interval variable lacks a true zero.

# Interval variables

- An interval variable is numeric and the intervals between values are equal.
- An interval variable lacks a true zero.
- “What was the maximum temperature in London over the past eight days?”

# Interval variables

| Measurement | Temperature (°C) |
|---|---|
| 1 | 17 |
| 2 | 17 |
| 3 | 14 |
| 4 | 14 |
| 5 | 15 |
| 6 | 16 |
| 7 | 15 |
| 8 | 12 |

# Ratio variables

- A ratio variable is numeric and the intervals between values are equal.
- A ratio variable has a true zero point, and zero has meaning.

# Ratio variables

- A ratio variable is numeric and the intervals between values are equal.
- A ratio variable has a true zero point, and zero has meaning.
- “How much do you bench?”

# Ratio variables

| Respondent | Bench press (kg) |
|---|---|
| 1 | 28 |
| 2 | 50 |
| 3 | 70 |
| 4 | 15 |
| 5 | 65 |
| 6 | 70 |
| 7 | 40 |
| 8 | 10 |

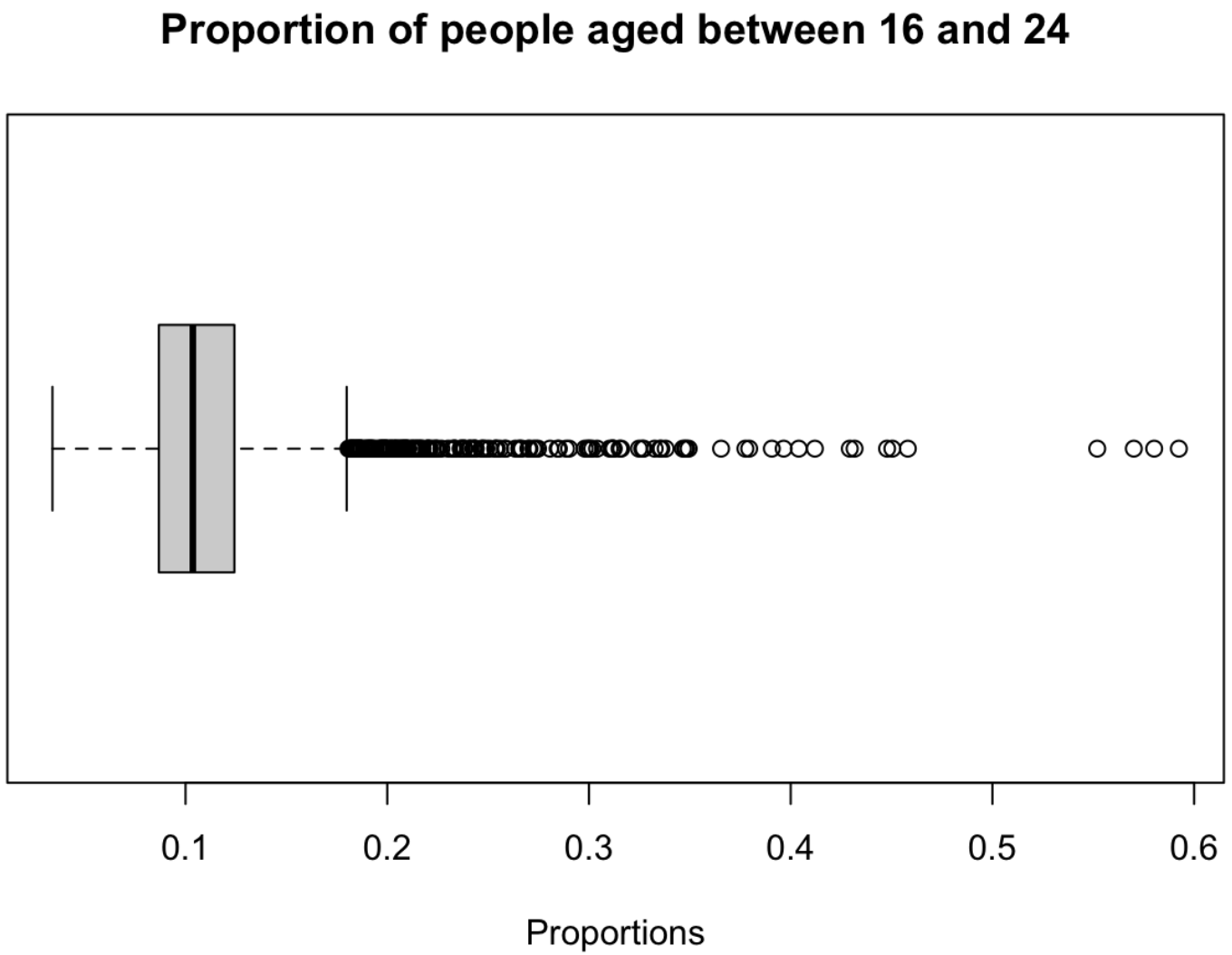
# Describing data

- Why do we care? Different variable types, different methods.
- Describing data is important to understand the data distribution, identify possible errors, decide on appropriate methods.
- How to describe?
  - Measures of central tendency.
  - Measures of dispersion.
  - Plots and maps.

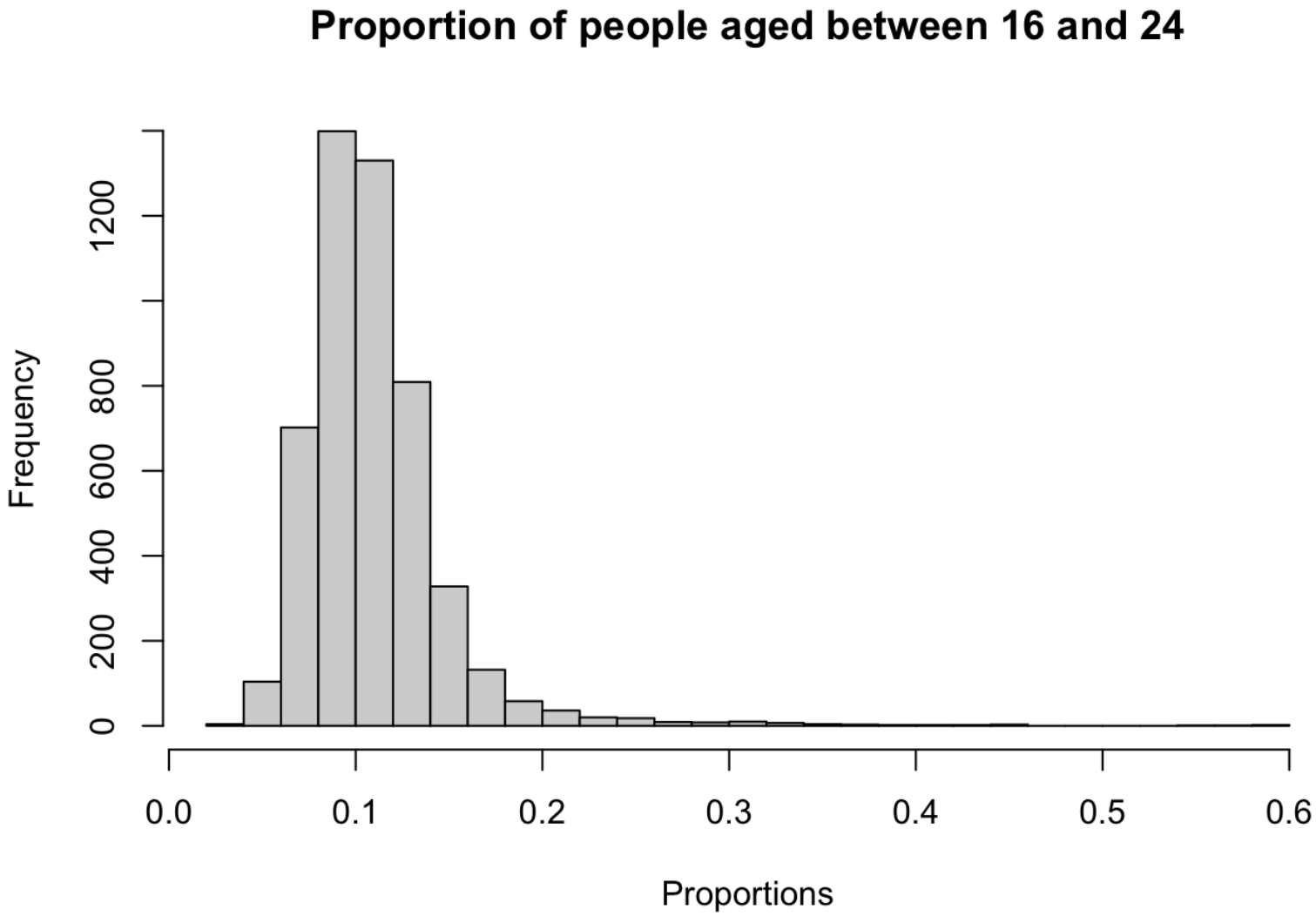
# Describing data

| Statistic | Definition |
| --- | --- |
| Mean | Sum of values divided by total number of cases. Vulnerable to extremes. |
| Median | Middle ranked observation. |
| Mode | Most common observation. |
| Range | Difference between lowest and highest value. |
| Interquartile range | Difference between the $25^{th}$ and $75^{th}$ percentile of values. |
| Standard deviation | Typical distance of values from the mean. |

# Describing data



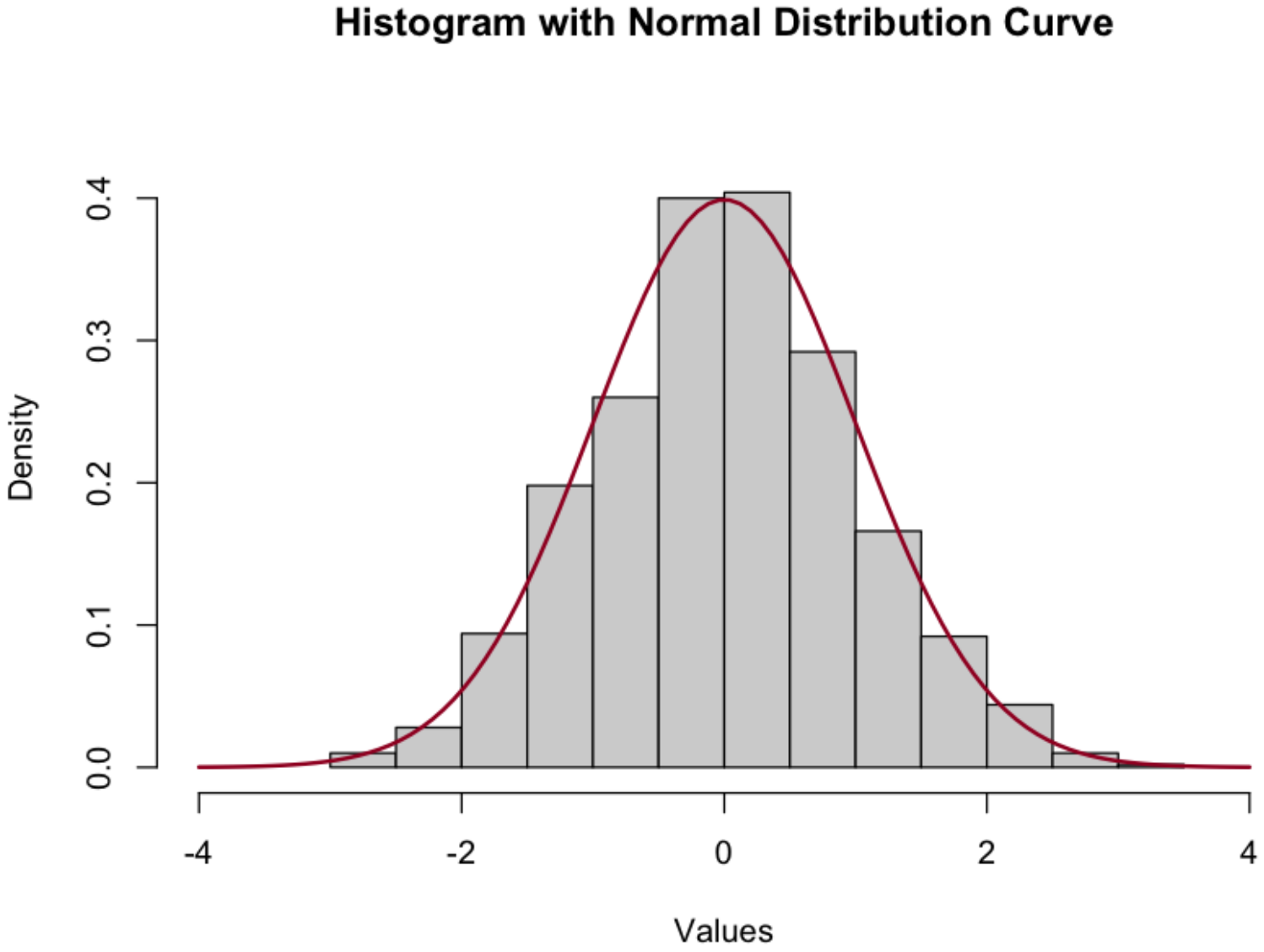
# Describing data



# Normal distribution

- The normal distribution is a symmetric, bell-shaped probability distribution with known properties.
- The mean, median, and mode of a normal distribution are equal.
- About 68% of the data falls within one standard deviation of the mean, 95% within two standard deviations.

# Normal distribution

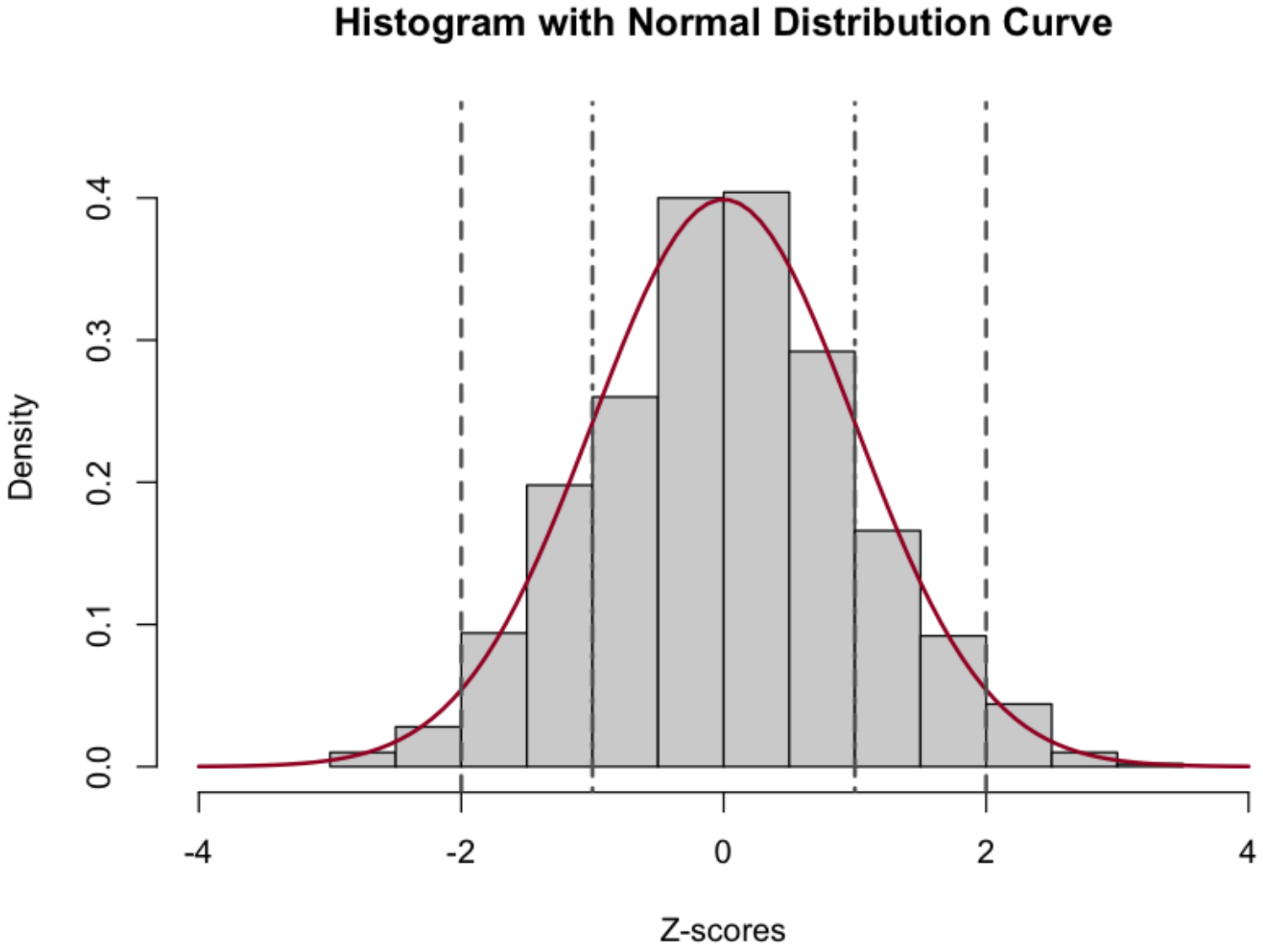


# Standard Normal distribution

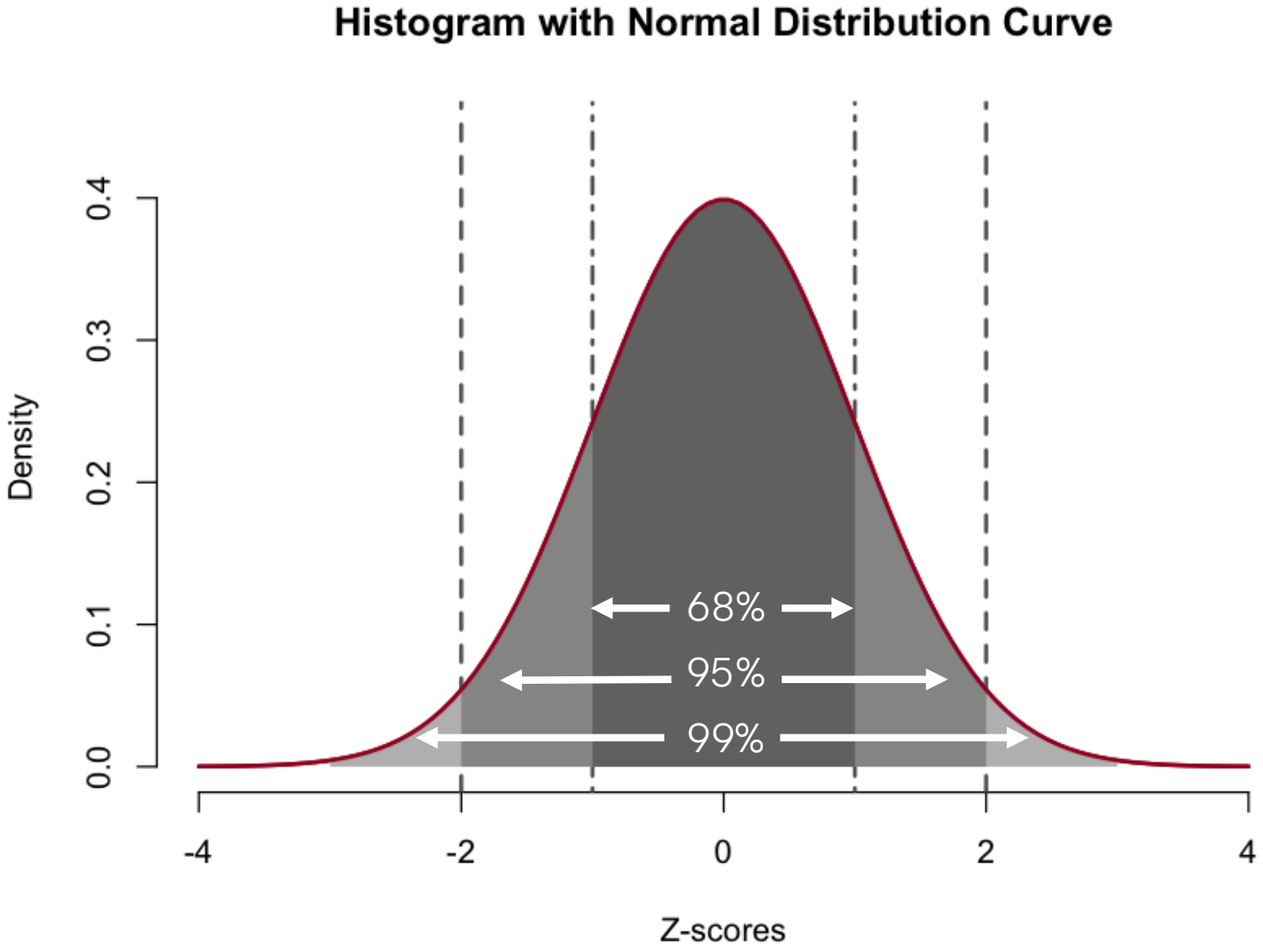
- Special case of a normal distribution: standard normal distribution.
- Standard normal distribution has a mean of 0 and a standard deviation of 1.
- All normal distributions can be transformed into this standard form using z-scores.
- A z-score represents the number of standard deviation units an observation is from the mean:

$$z = \frac{x - \mu}{\sigma}$$

# Normal distribution



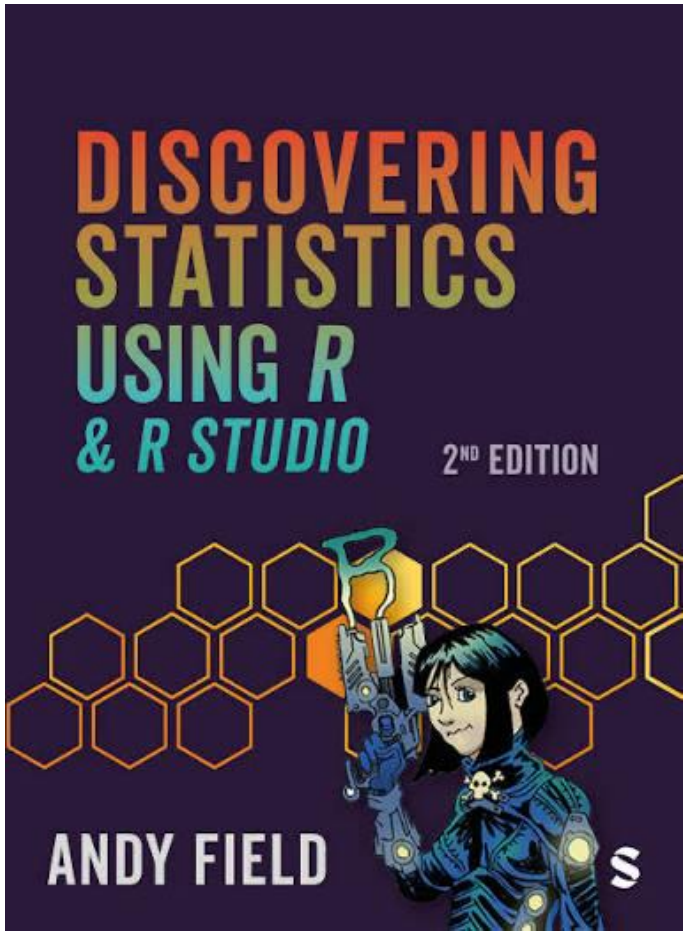
# Normal distribution



# Normal distribution

- Some variables are approximately normal (e.g. height). Many in human geography are not: income, house prices and population density are skewed.
- Because the properties are known, the mean and standard deviation tell us how likely values are (z-scores).
- Many statistical tests assume normally distributed data, so we look at the distribution first.
- Next week: sample means are also approximately normal. This lets us estimate how far a sample mean is from the population mean, and test hypotheses.

# Suggested reading



# R for data analysis

# Programming languages

- Consist of a formal set of instructions in the form of code that you can use to write software or perform computational tasks.
- Allow for custom solutions, automation, and complex operations.

# Programming languages

- Identifiers and primitive data types.
- Assignment, arithmetic, logical and relational operators.
- Expression and statements, debugging.
- Flow of control: selection and repetition.
- Functions, parameters passing, call by value and reference.
- Object-oriented programming.
- 1/2 dimensional arrays, strings and data structures.

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# Programming languages

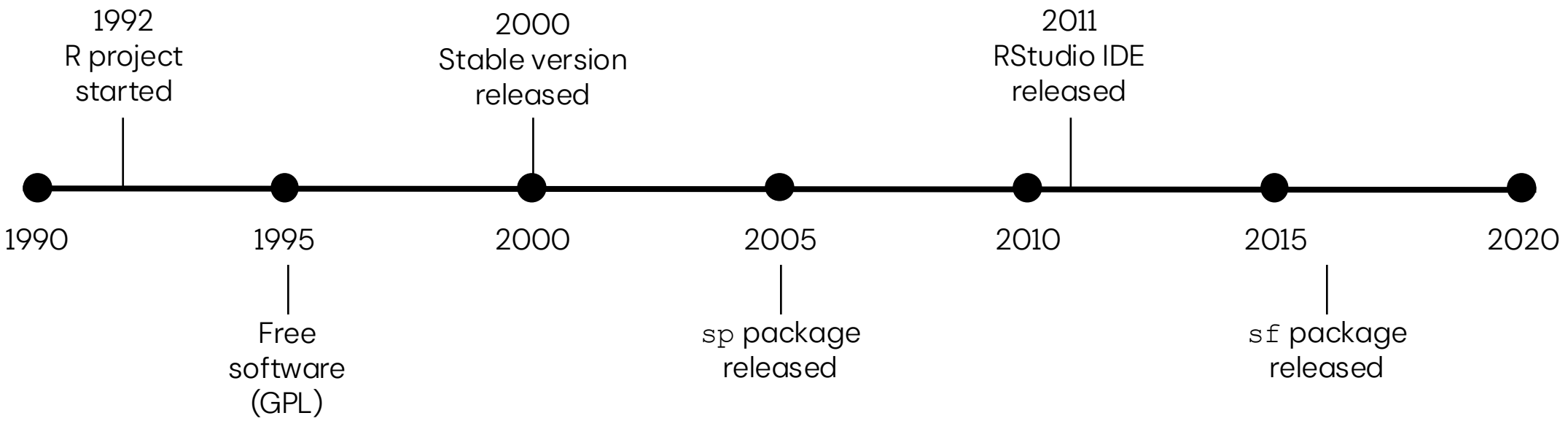
- Ιδεντιφιερς ανδ πριμιτιφε δατα τψπες.
- Ασσιγνμεντ, αριτημετιξ, λογιξαλ ανδ ρελατιοναλ οπερατορς.
- Εχπρεσσιον ανδ στατεμεντς, δεβυγγινγ.
- Φλοω οφ ξοντρολ· σελεξτιον ανδ ρεπετιτιον.
- Φυνξτιονς, παραμετερς πασσινγ, ξαλλ βψ αφλυε ανδ ρεφερενξε.
- Οβσεξτ−οριεντεδ προγραμμινγ.
- 1/2 διμενσιοναλ αρραψς, στρινγς ανδ δατα στρυξτυρες.

# The absolute basics

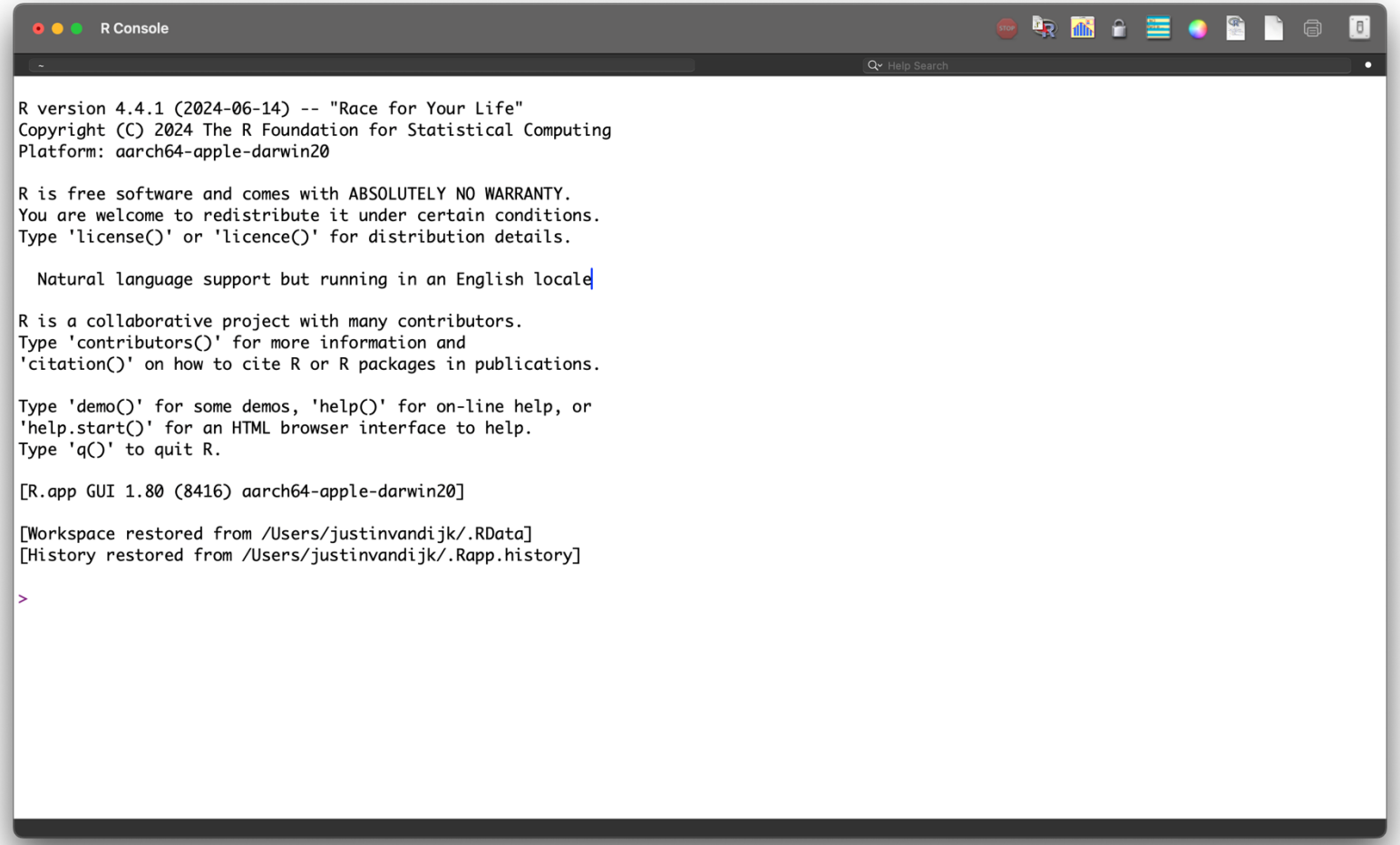
- R is primarily used through interactive command-line interfaces and scripts.
- R is the programming language, but we can interact with it using other software.
- R efficiently handles various data structures, particularly vectors and tables.
- R's functionality can be extended through a vast ecosystem of packages.
- R is free and open source.

# A little history

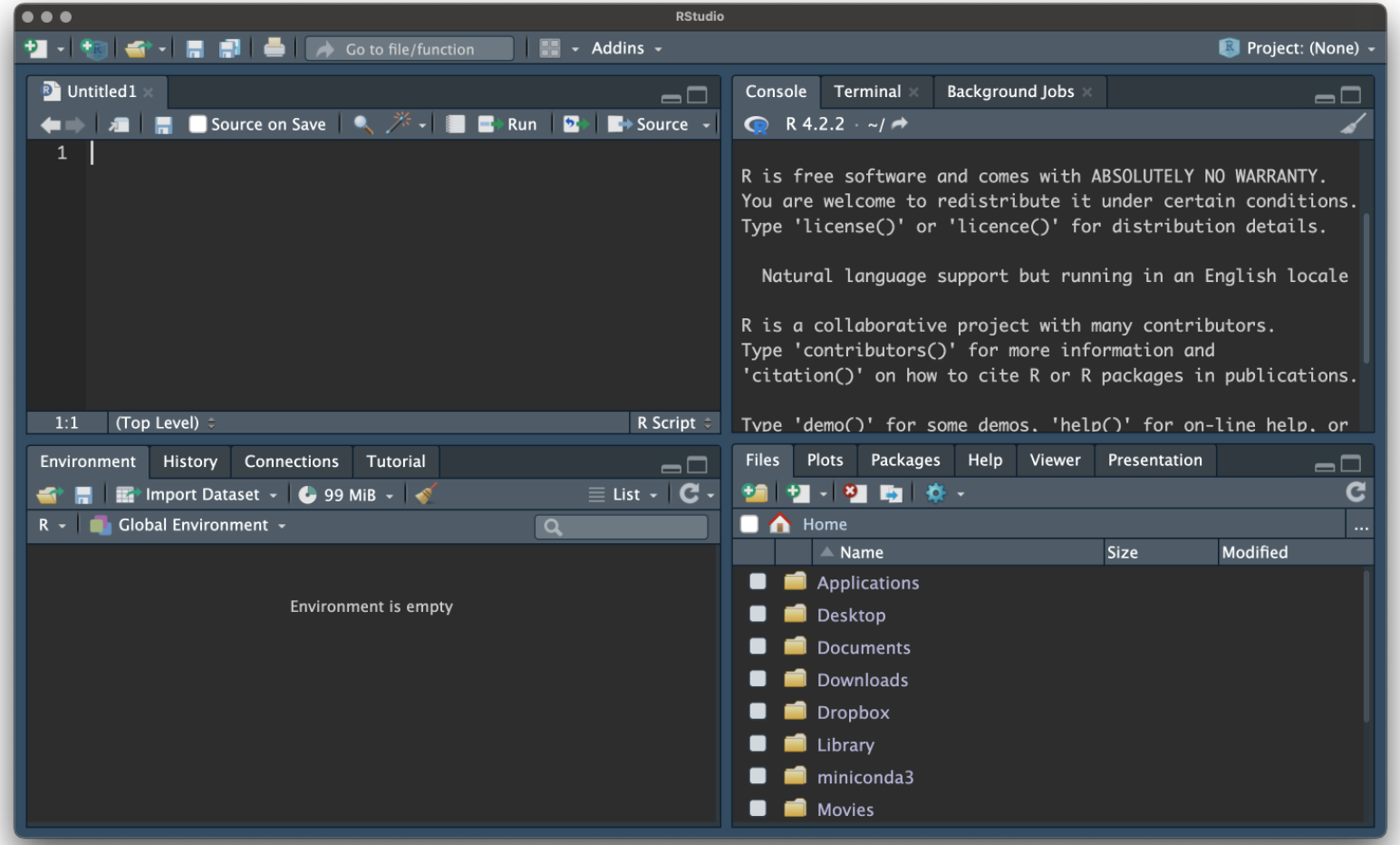
- R is a programming language originally developed for statistical purposes.
- 1975: Bell Labs develops a language for Statistical Analysis ("S").
- 1992: Ross Ihaka and Robert Gentleman start R, an implementation of "S".



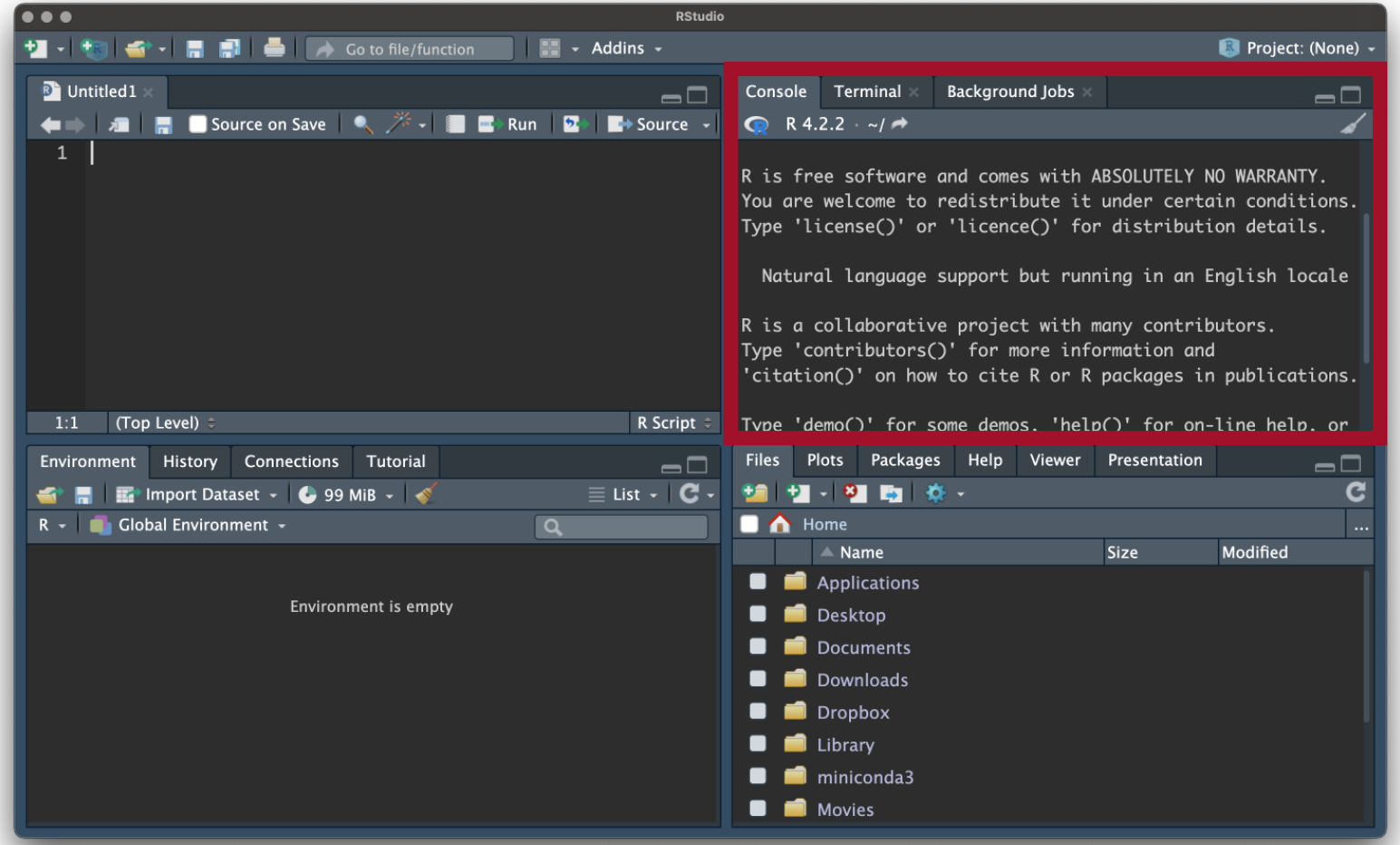
# Working with R



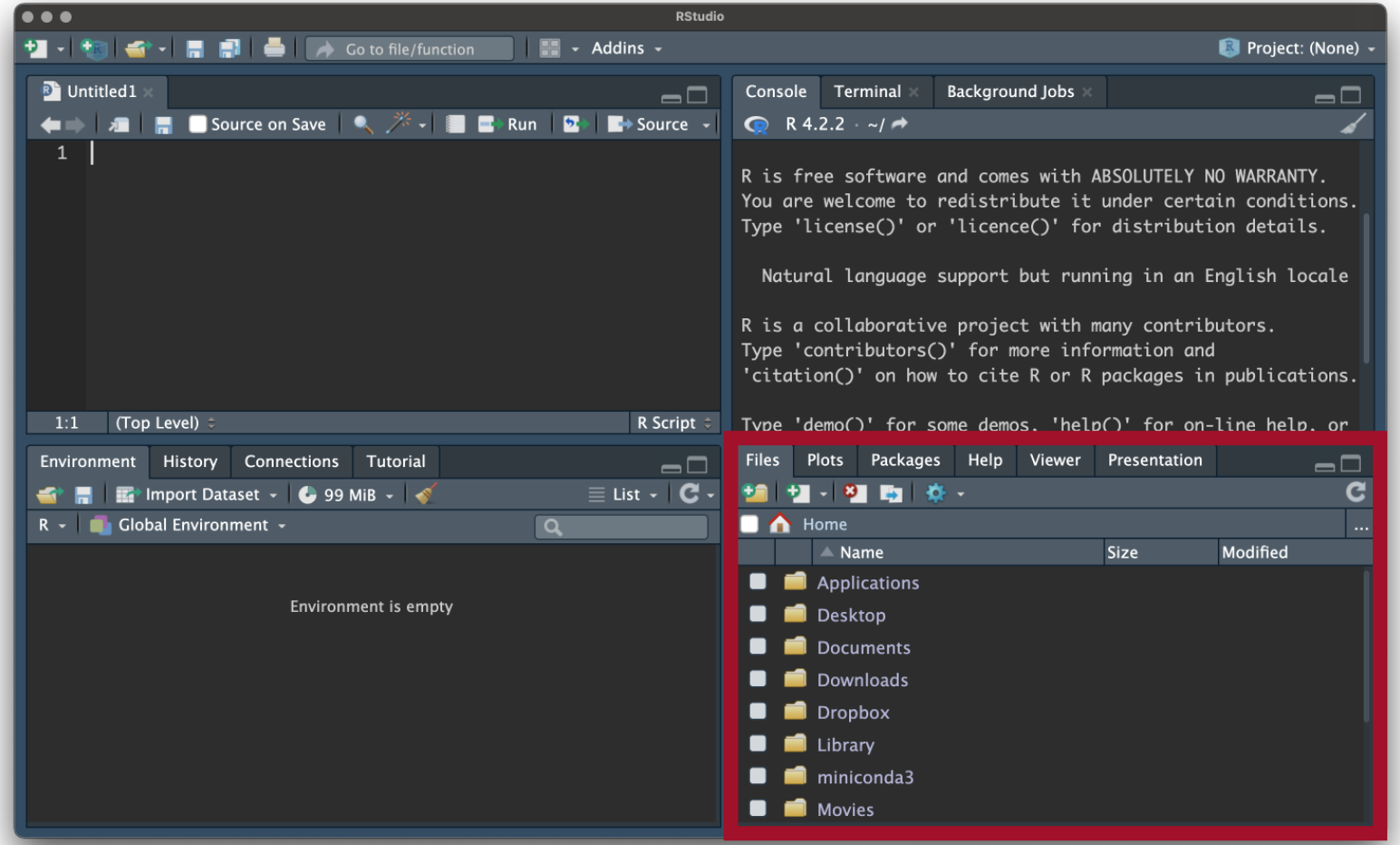
# Working with R



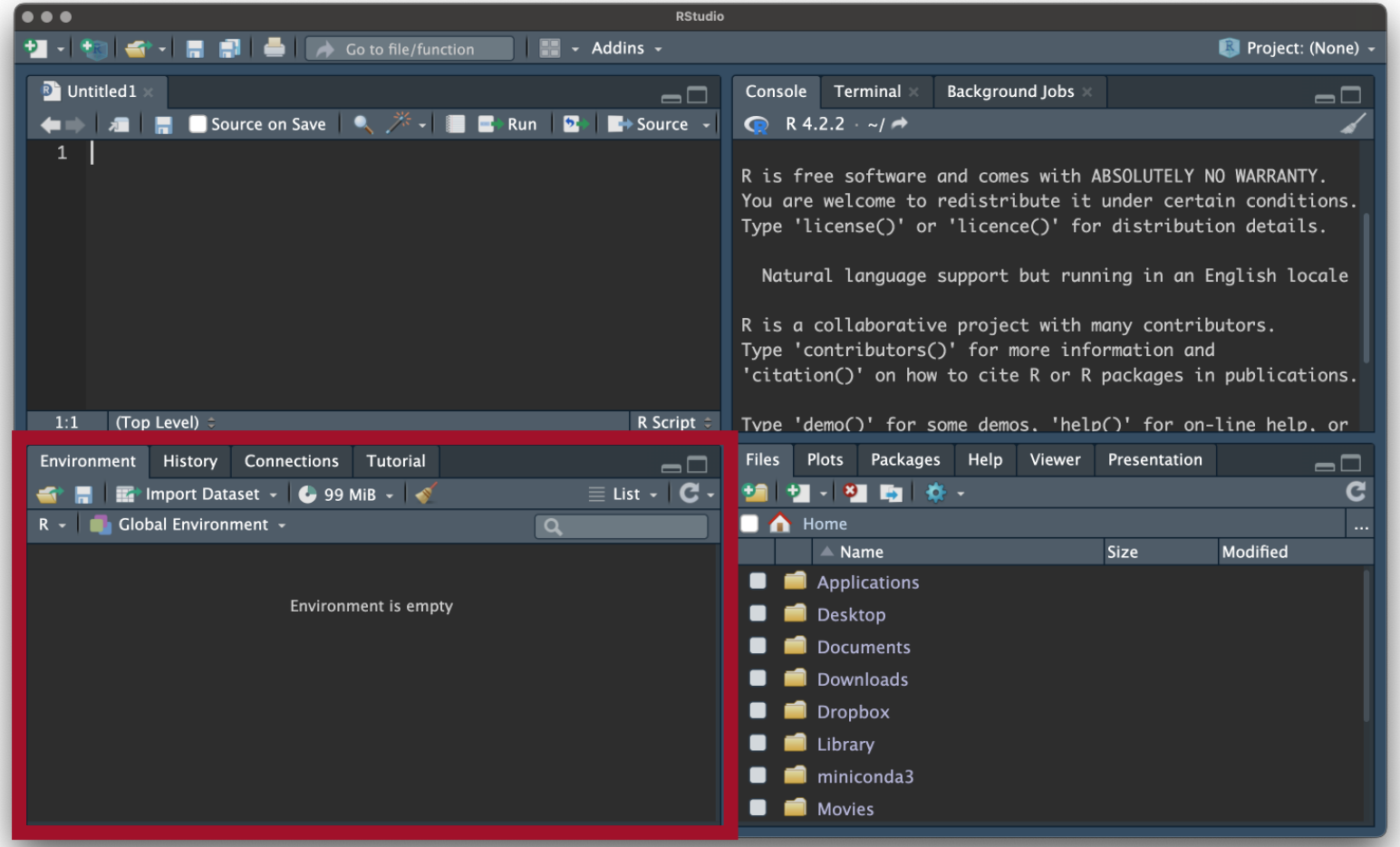
# Working with R



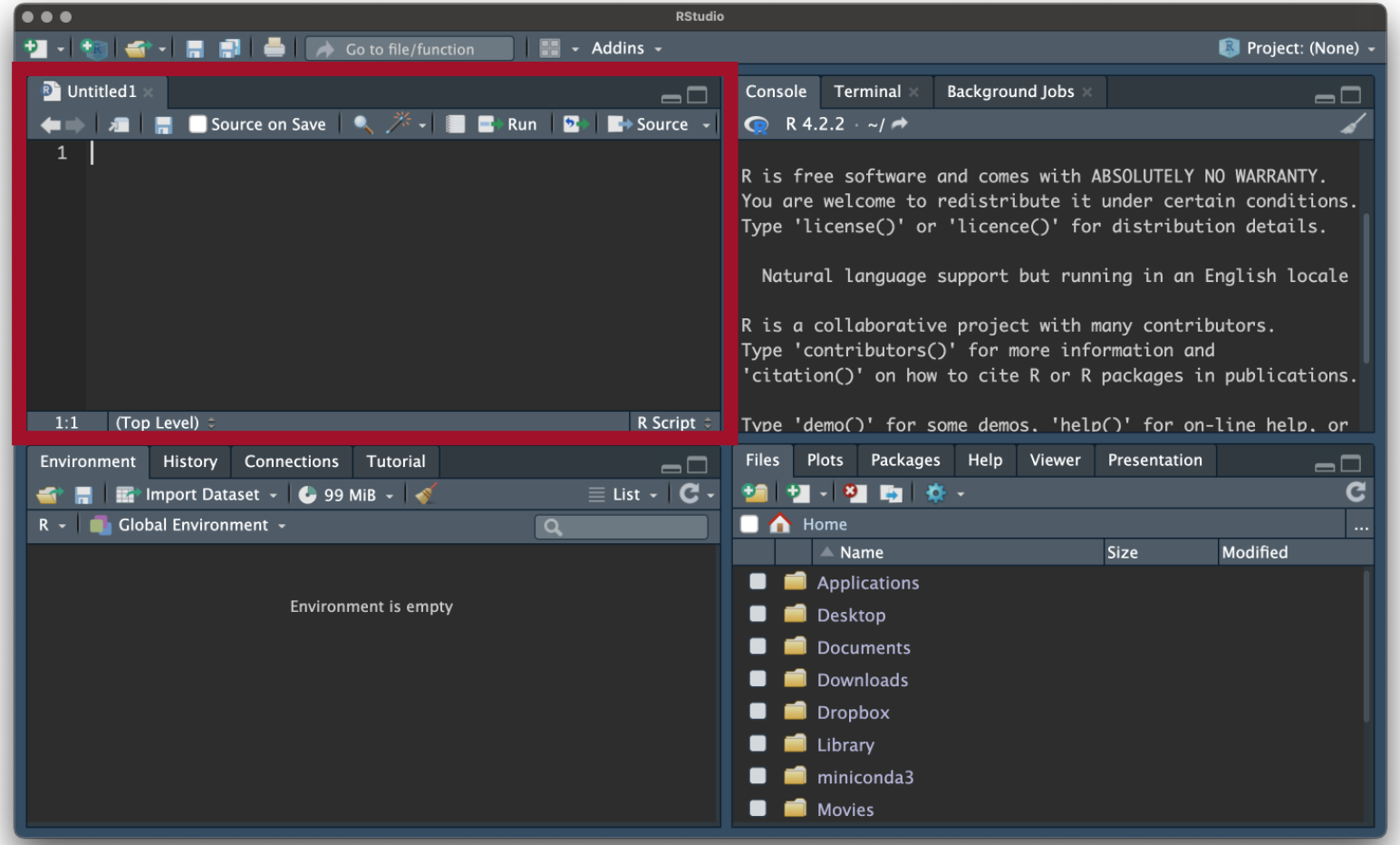
# Working with R



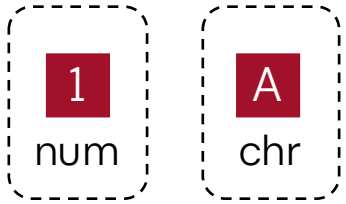
# Working with R



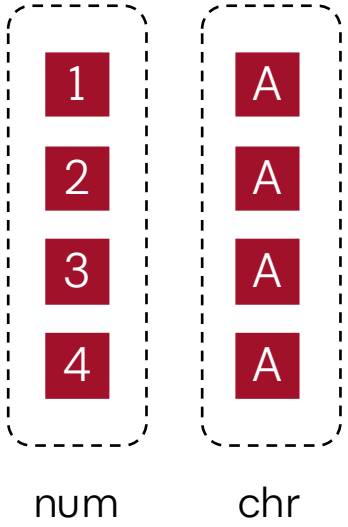
# Working with R



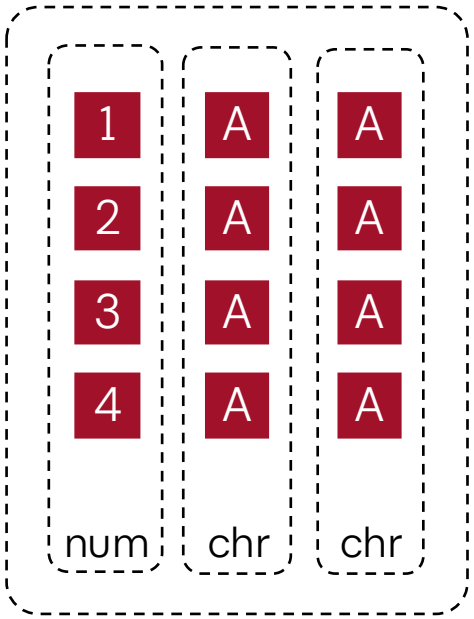
# Principles of R: Data structures



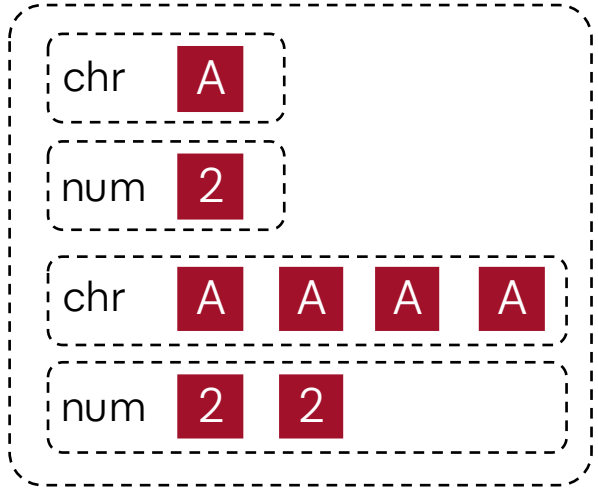

Scalar




Vector




Dataframe




List

# Principles of R: Objects

- All scalars, vectors, tables, and lists can be assigned to a name, which creates an object.
- Objects are used to store information to be referenced and manipulated in a computer program.

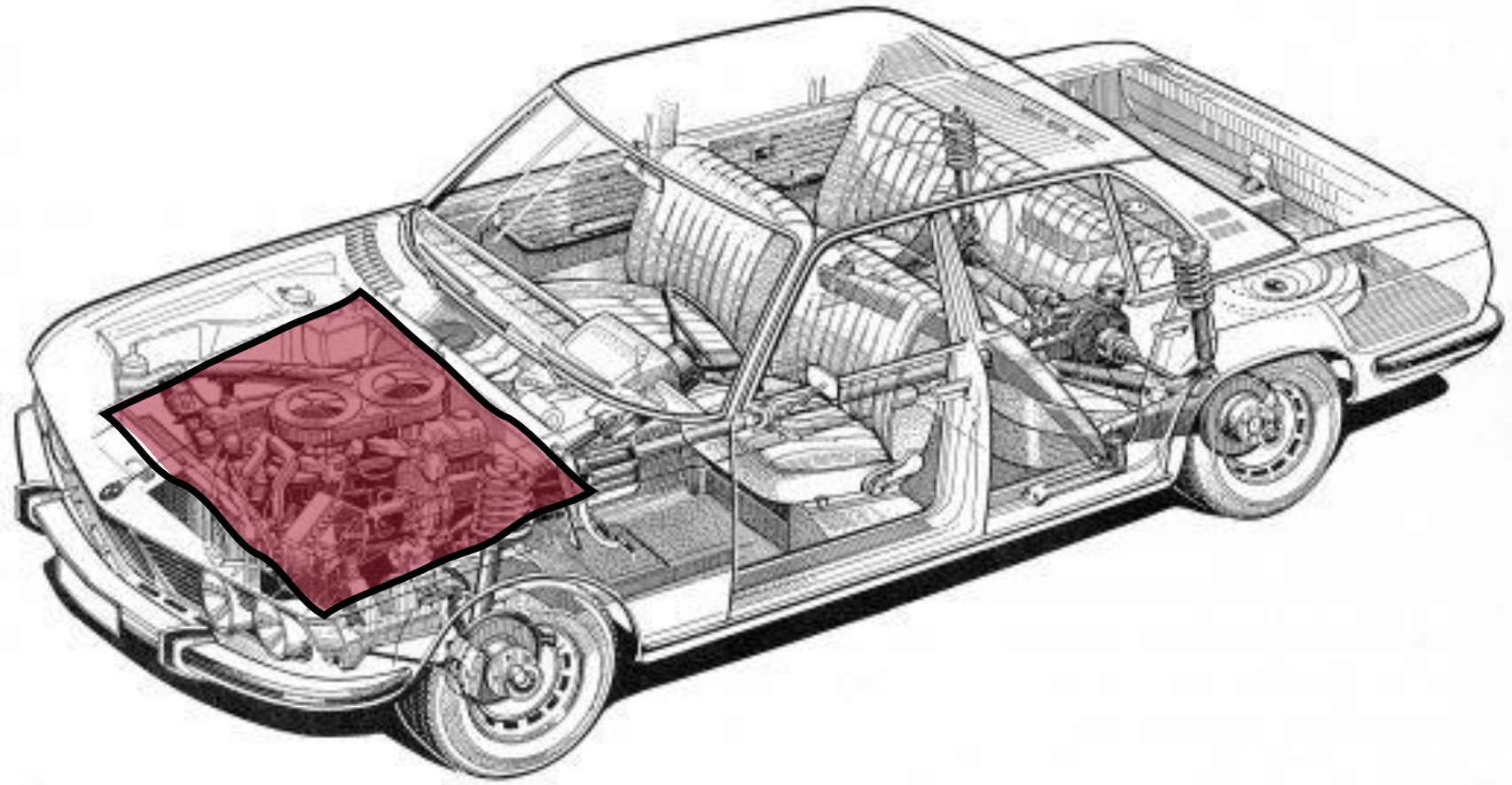
# Principles of R: Functions

- Objects can be used as an input for functions.
- Functions are pieces of code designed to accomplish specific tasks.
- Once a function is written, it can be reused.

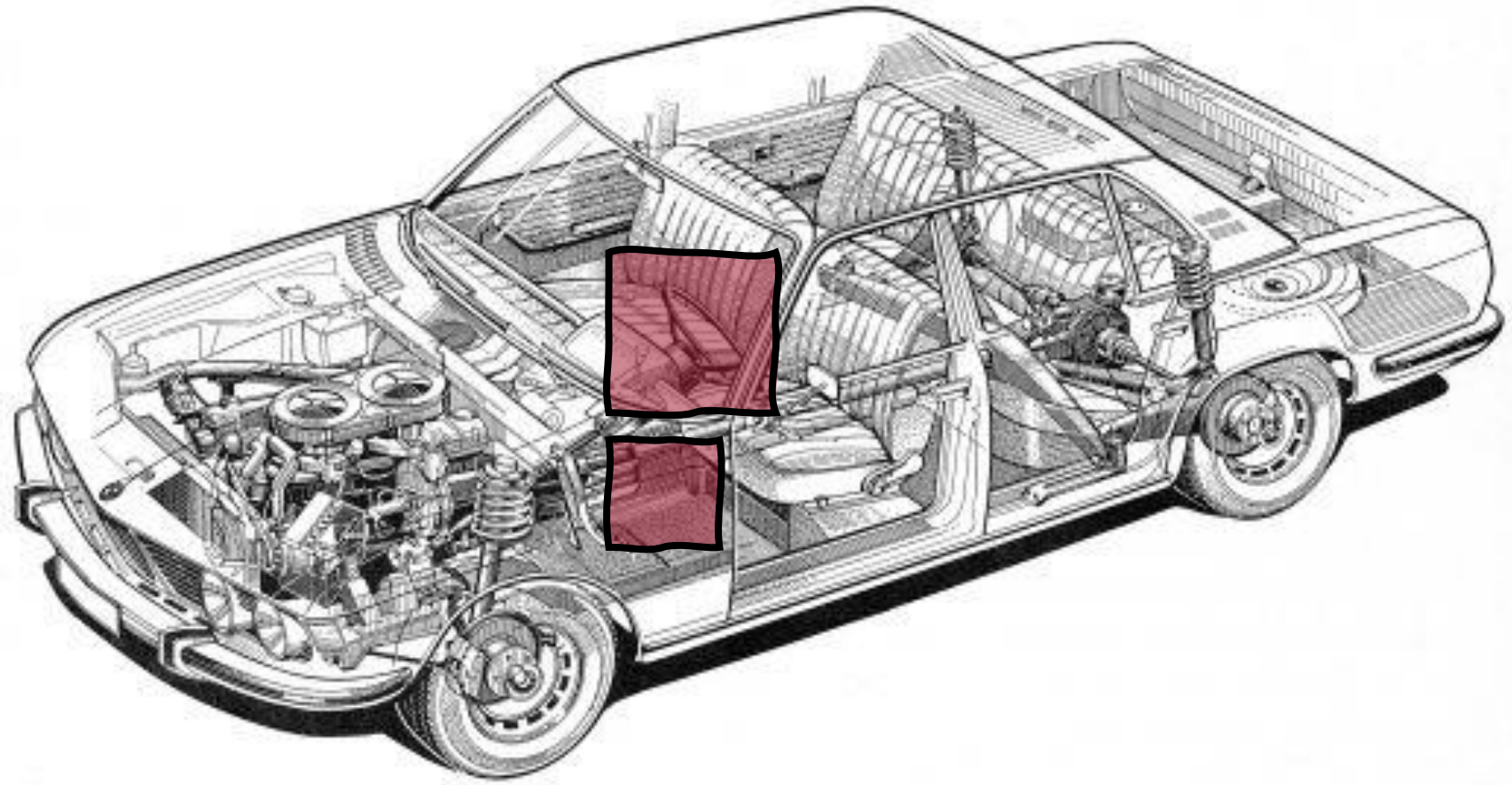
# When R gives an error

- Error messages are normal: read the message and the line it names.
- Most errors are typos, missing brackets or code that was not run.
- '`object not found`': run the code that creates it, check the spelling.
- '`could not find function`': load the package with `library()`.
- '`there is no package called`': install it with `install.packages()`.
- A + instead of > in the console: press Esc and try again.

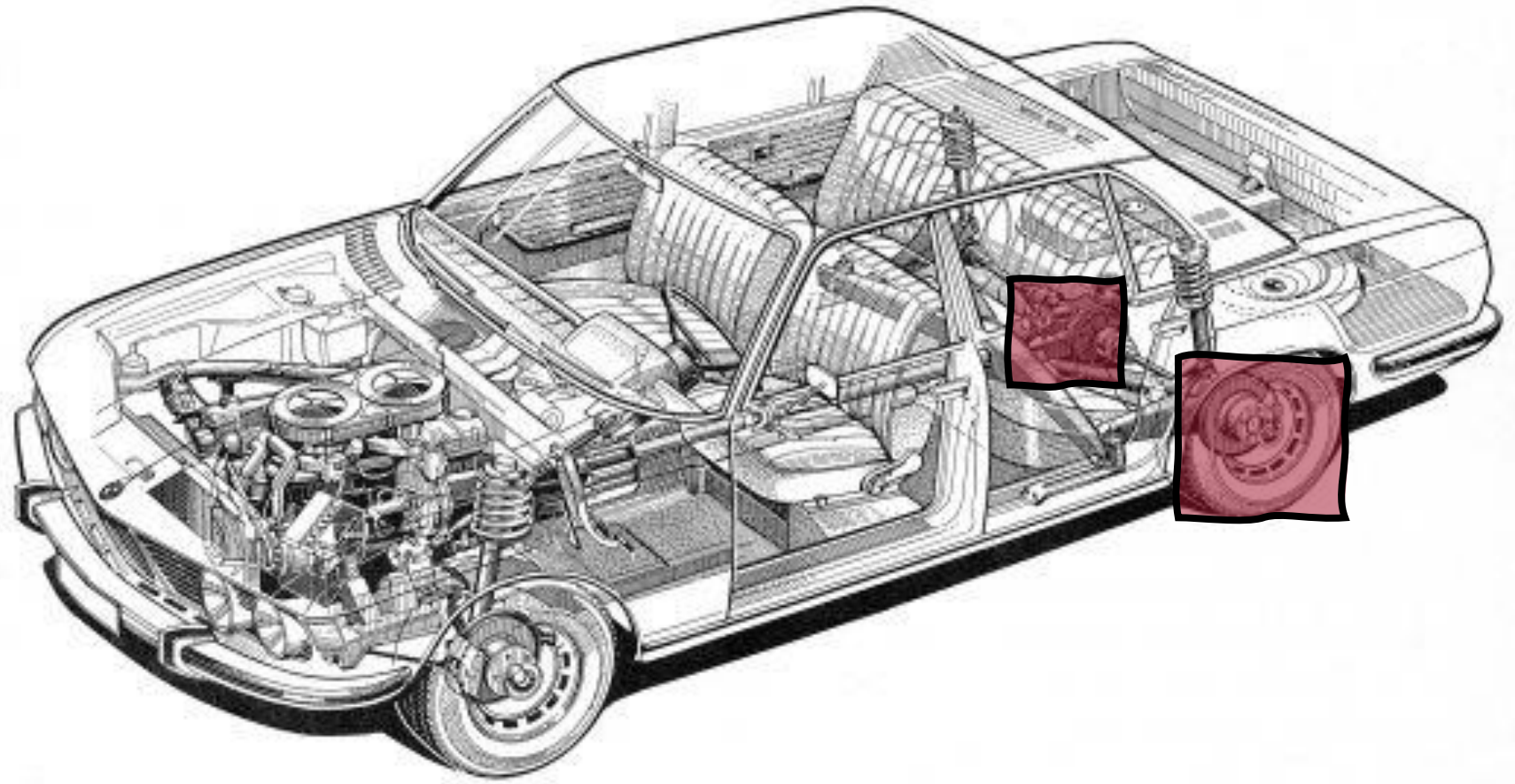
# Principles of R: R and RStudio



# Principles of R: R and RStudio



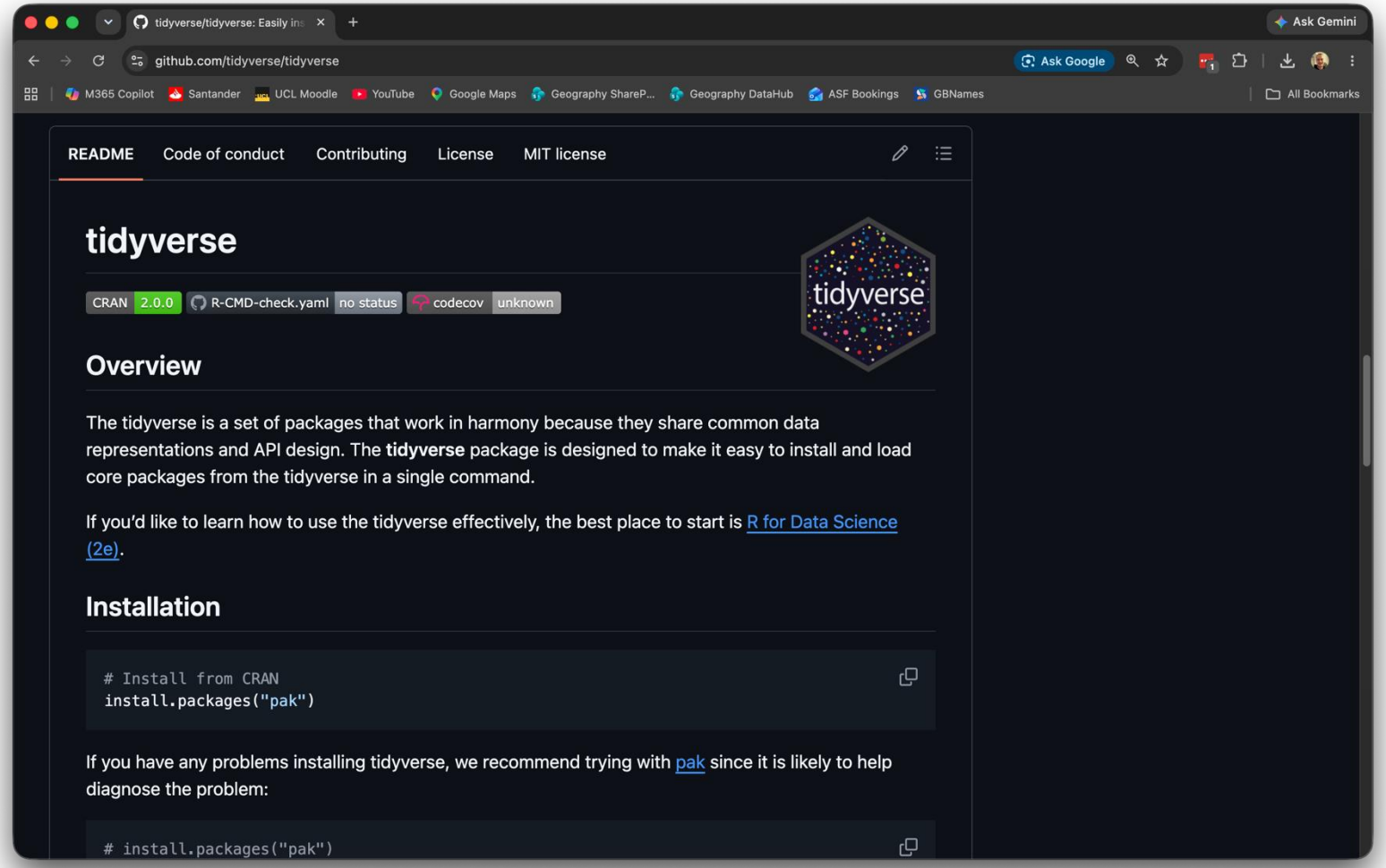
# Principles of R: Extensions



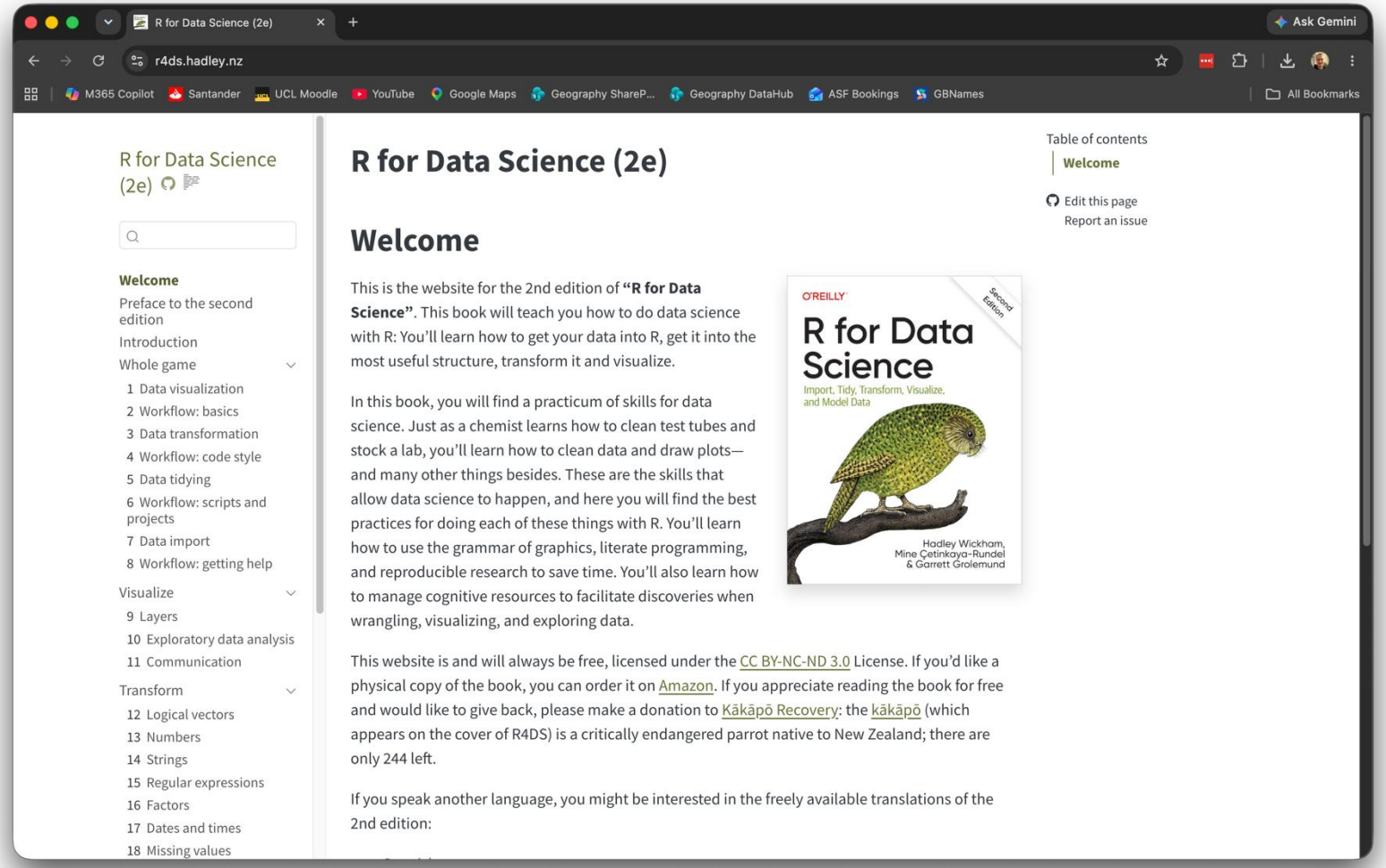
# Principles of R: Extensions

- Available on The Comprehensive R Archive Network: CRAN
- CRAN is a repository for R packages and software: the ‘shop’ for our add-ons.
- Currently CRAN features over 25,600 available packages (02/11/2026).
- Allows developers to submit and maintain their own R packages.

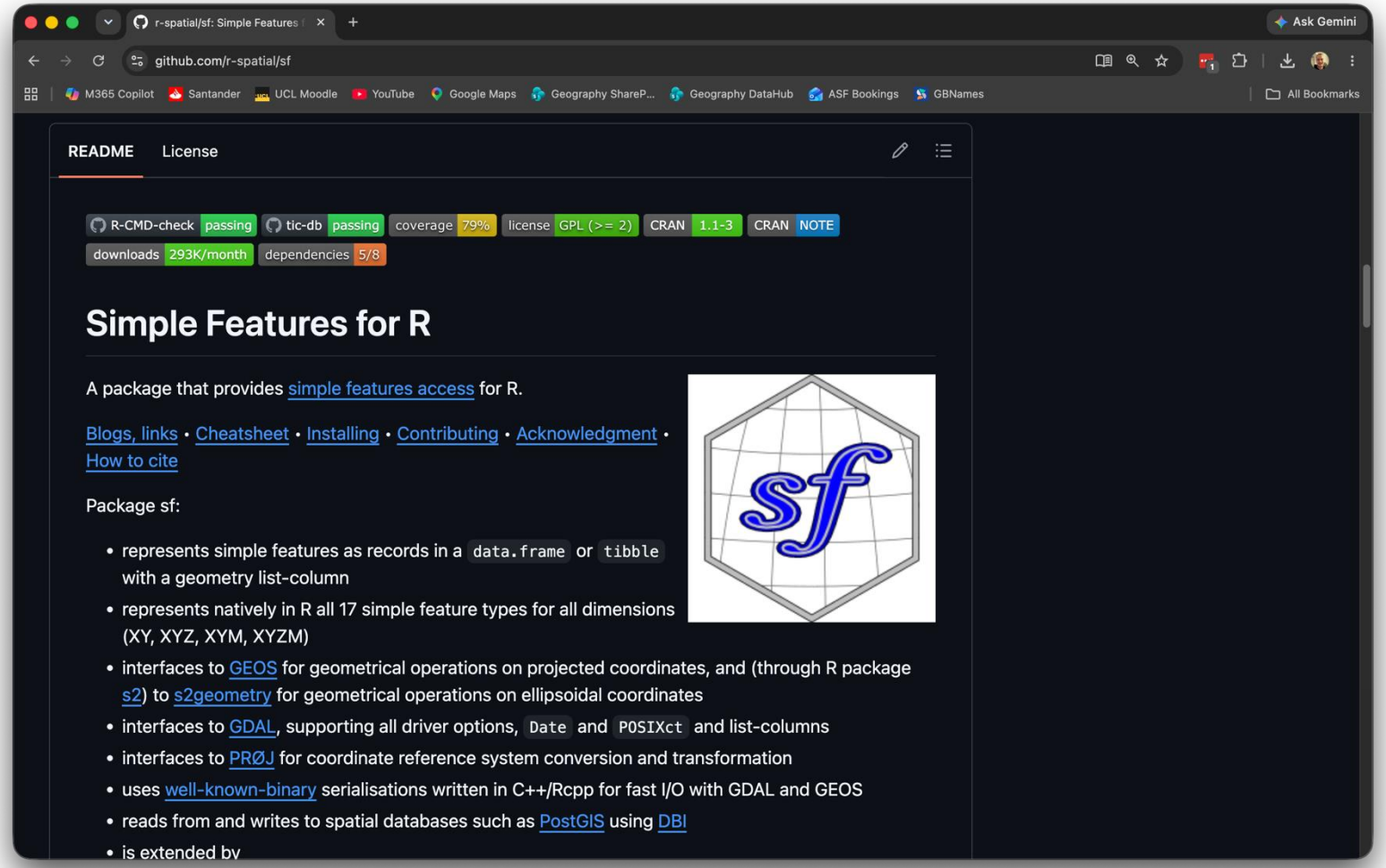
# tidyverse


tidyverse/tidyverse: Easily ins
Ask Gemini
github.com/tidyverse/tidyverse
Ask Google
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
README
Code of conduct
Contributing
License
MIT license
tidyverse
CRAN 2.0.0
R-CMD-check.yaml no status
codecov unknown
tidyverse
Overview
The tidyverse is a set of packages that work in harmony because they share common data representations and API design. The tidyverse package is designed to make it easy to install and load core packages from the tidyverse in a single command.
If you'd like to learn how to use the tidyverse effectively, the best place to start is R for Data Science (2e).
Installation
# Install from CRAN
install.packages("pak")
If you have any problems installing tidyverse, we recommend trying with pak since it is likely to help diagnose the problem:
# install.packages("pak")

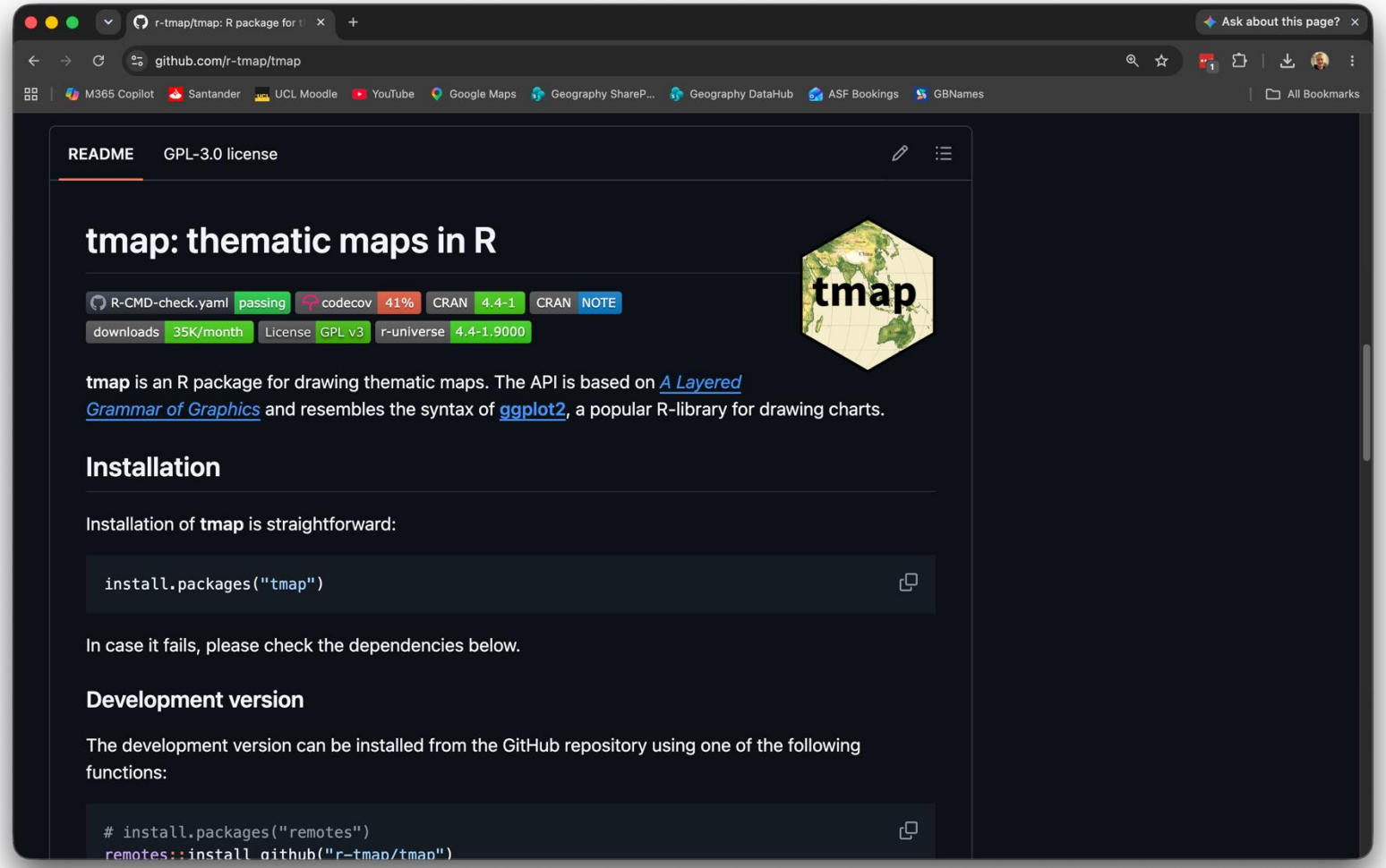
# tidyverse


R for Data Science (2e)
Ask Gemini
r4ds.hadley.nz
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
R for Data Science (2e)
Welcome
Preface to the second edition
Introduction
Whole game
1 Data visualization
2 Workflow: basics
3 Data transformation
4 Workflow: code style
5 Data tidying
6 Workflow: scripts and projects
7 Data import
8 Workflow: getting help
Visualize
9 Layers
10 Exploratory data analysis
11 Communication
Transform
12 Logical vectors
13 Numbers
14 Strings
15 Regular expressions
16 Factors
17 Dates and times
18 Missing values
R for Data Science (2e)
Welcome
This is the website for the 2nd edition of "R for Data Science". This book will teach you how to do data science with R: You'll learn how to get your data into R, get it into the most useful structure, transform it and visualize.
In this book, you will find a practicum of skills for data science. Just as a chemist learns how to clean test tubes and stock a lab, you'll learn how to clean data and draw plots—and many other things besides. These are the skills that allow data science to happen, and here you will find the best practices for doing each of these things with R. You'll learn how to use the grammar of graphics, literate programming, and reproducible research to save time. You'll also learn how to manage cognitive resources to facilitate discoveries when wrangling, visualizing, and exploring data.
O'REILLY
Second Edition
R for Data Science
Import, Tidy, Transform, Visualize, and Model Data
Hadley Wickham, Mine Çetinkaya-Rundel & Garrett Grolemund
This website is and will always be free, licensed under the CC BY-NC-ND 3.0 License. If you'd like a physical copy of the book, you can order it on Amazon. If you appreciate reading the book for free and would like to give back, please make a donation to Kākāpō Recovery: the kākāpō (which appears on the cover of R4DS) is a critically endangered parrot native to New Zealand; there are only 244 left.
If you speak another language, you might be interested in the freely available translations of the 2nd edition:
Table of contents
Welcome
Edit this page
Report an issue

# Simple Features



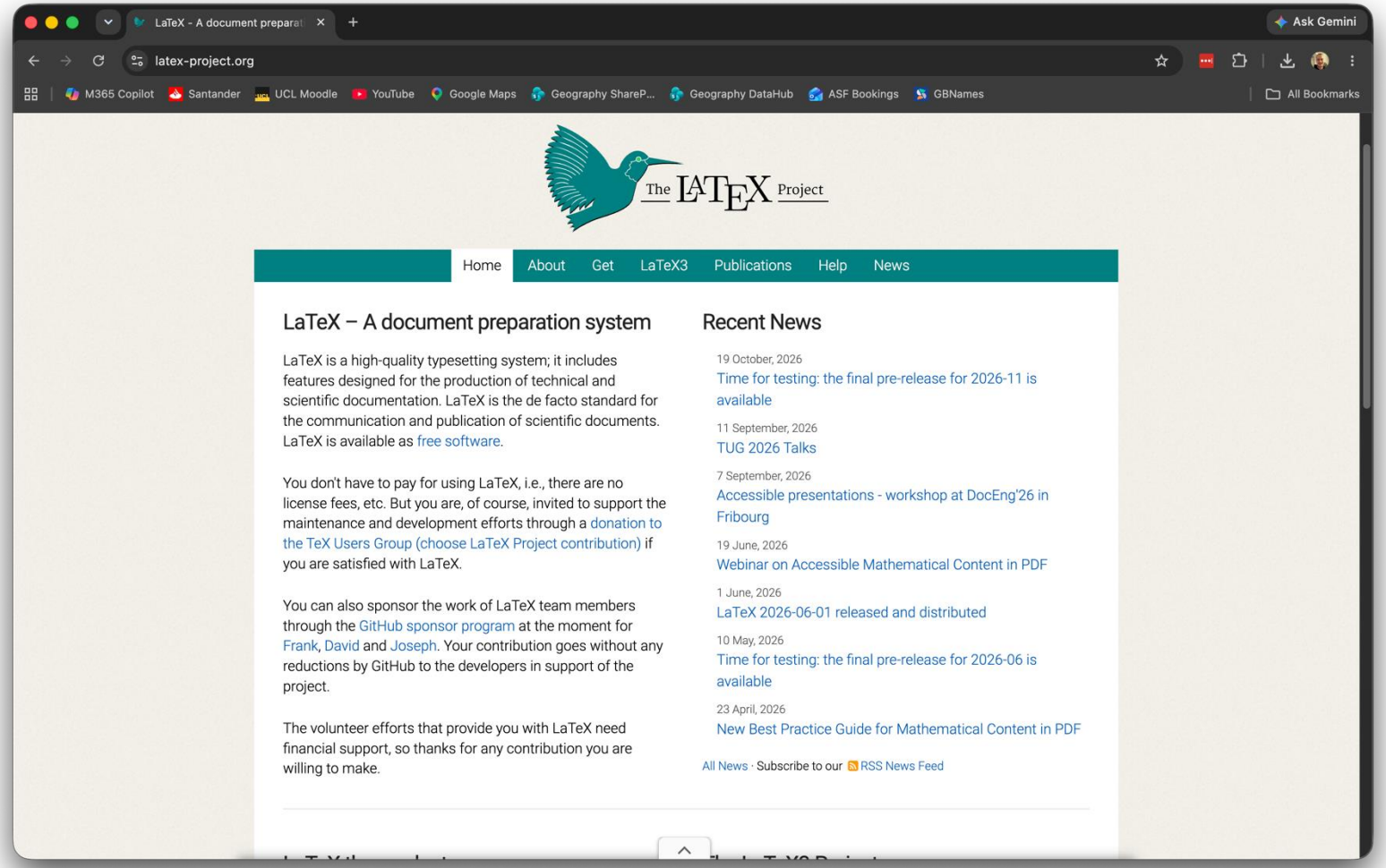
# tmap


r-tmap/tmap: R package for t
Ask about this page?
github.com/r-tmap/tmap
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
README
GPL-3.0 license
tmap: thematic maps in R
R-CMD-check.yaml passing
codecov 41%
CRAN 4.4-1
CRAN NOTE
downloads 35K/month
License GPL v3
r-universe 4.4-1.9000
tmap
tmap is an R package for drawing thematic maps. The API is based on A Layered Grammar of Graphics and resembles the syntax of ggplot2, a popular R-library for drawing charts.
Installation
Installation of tmap is straightforward:
install.packages("tmap")
In case it fails, please check the dependencies below.
Development version
The development version can be installed from the GitHub repository using one of the following functions:
# install.packages("remotes")
remotes::install_github("r-tmap/tmap")

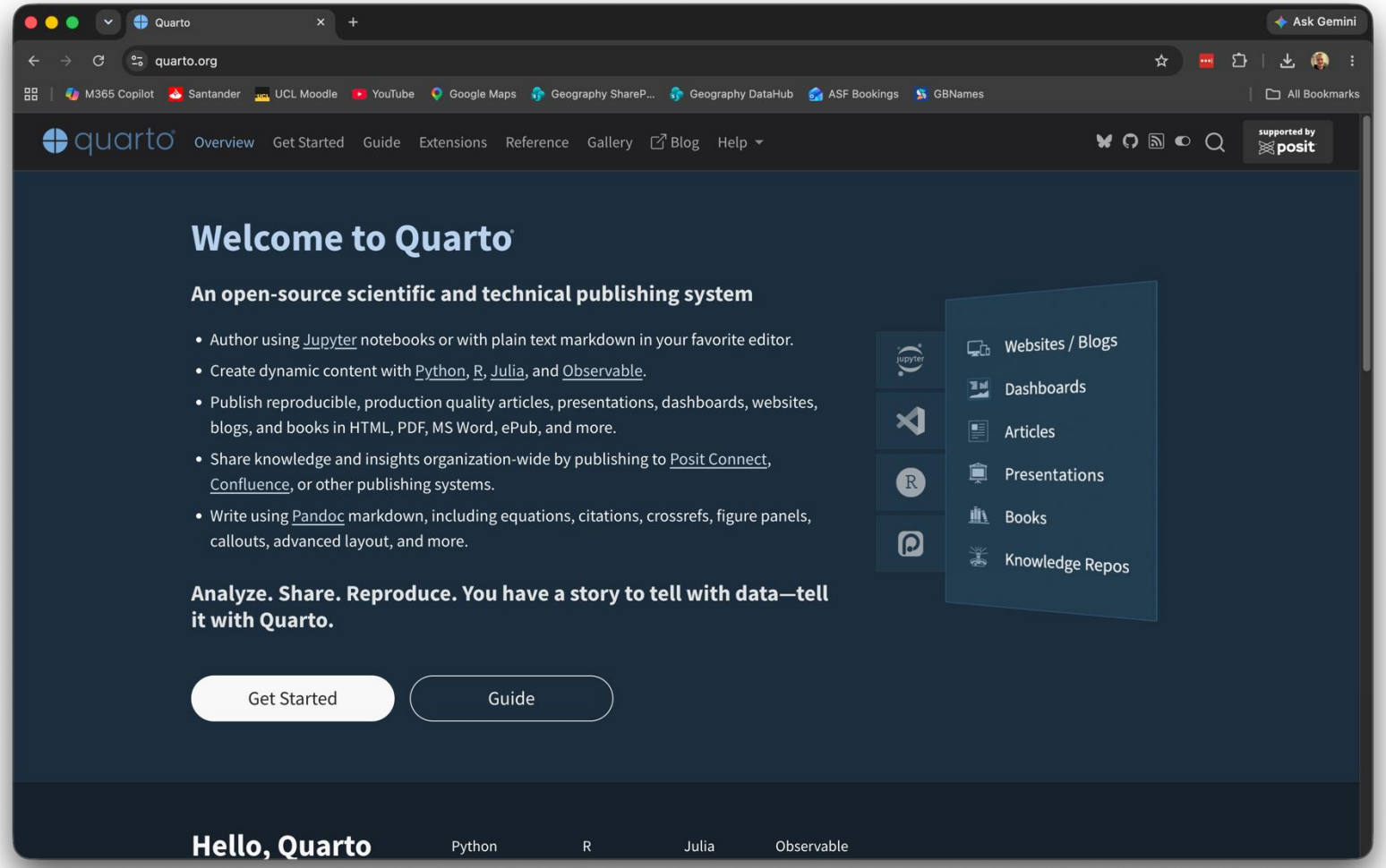
# Reproducibility with Markdown and Quarto

- Markdown is a lightweight markup language designed for creating formatted text using a plain-text editor. It allows you to write in a simple, readable format that can be easily converted to HTML, PDF, or other formats.
- Quarto extends Markdown by integrating code, such as R, with the text. This allows you to: embed code, execute code, create dynamic reports, format and typeset your content.

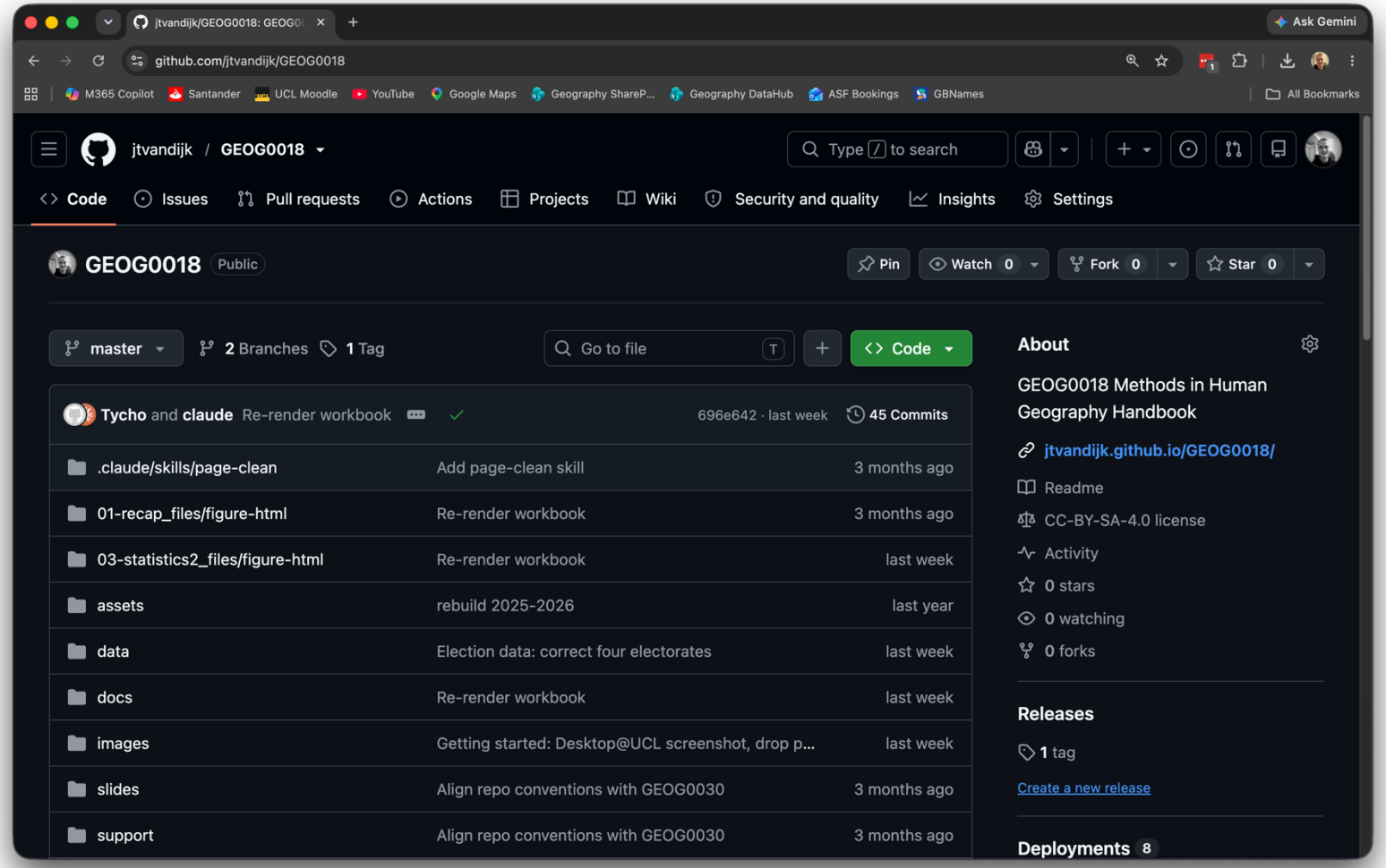
# Reproducibility with Markdown and Quarto



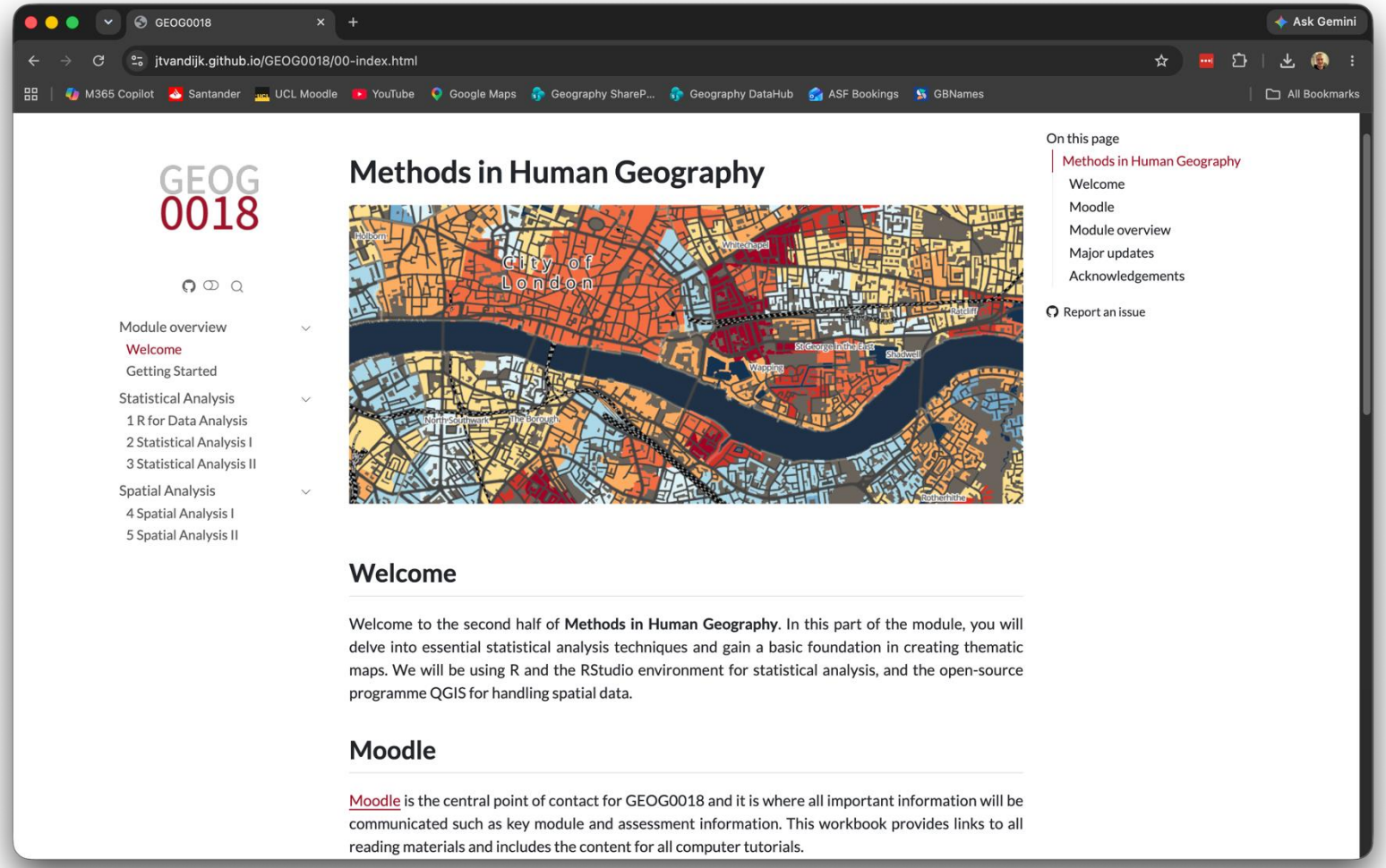
# Reproducibility with Markdown and Quarto



# Reproducibility with Markdown and Quarto



# Reproducibility with Markdown and Quarto

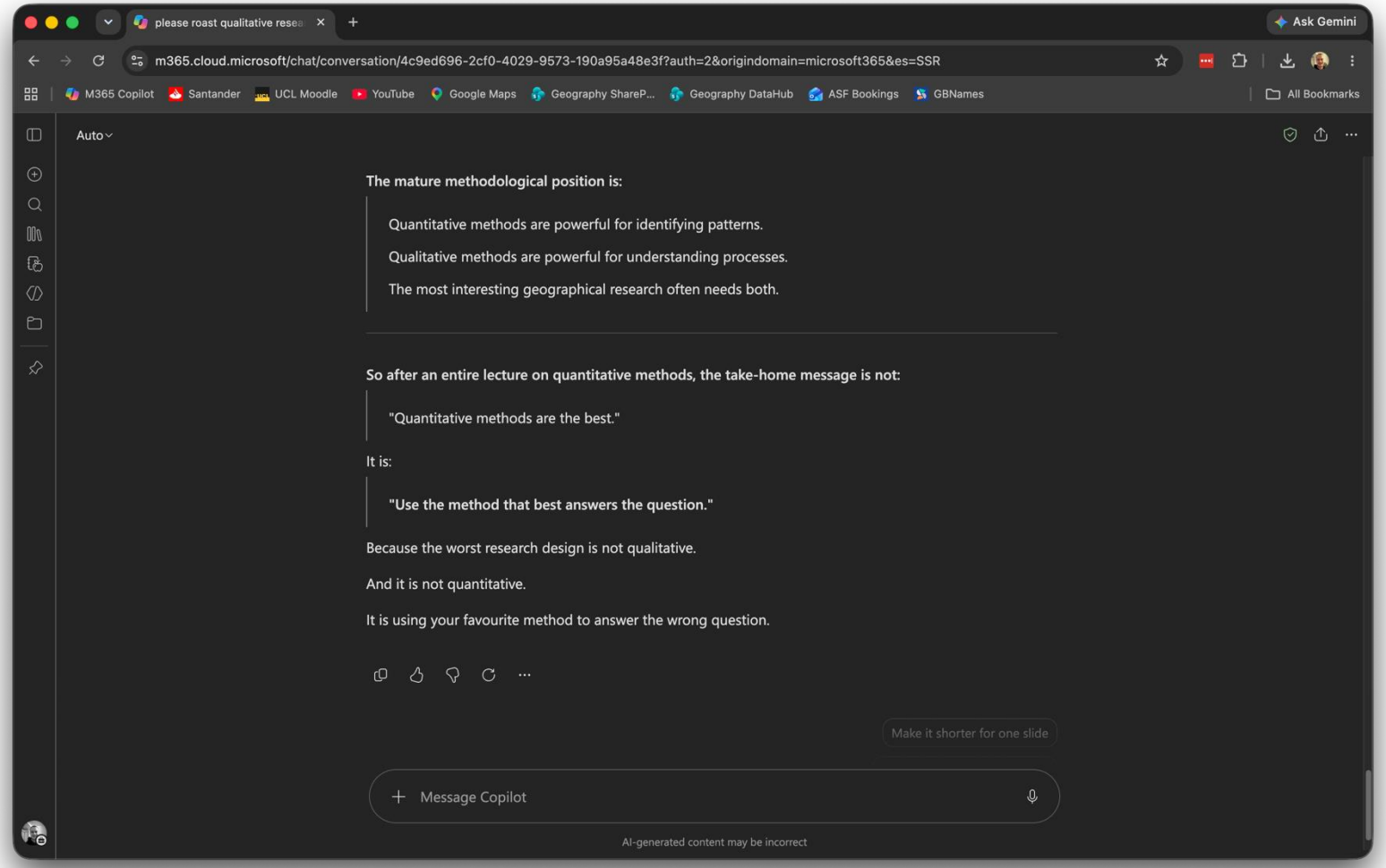


# Summary

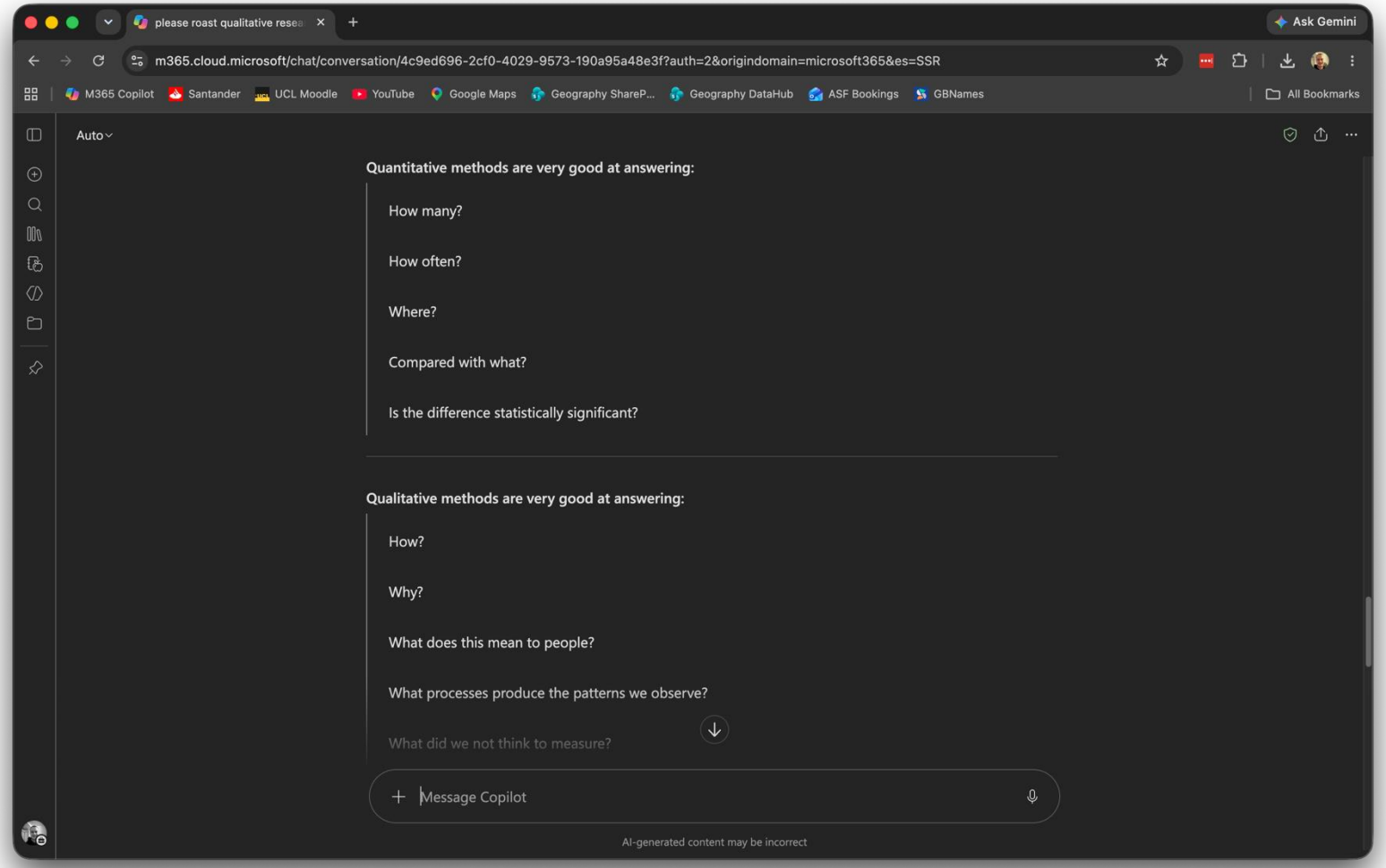
# Summary

- Quantitative methods focus on the collection and analysis of numerical data to identify patterns, test hypotheses, and make predictions.
- When designing research questions, consider how they align with the data you will collect and the statistical methods you will use.
- Understanding variable types (nominal, ordinal, interval, ratio) and their distributions is essential for selecting appropriate analysis techniques.
- R is a powerful tool for conducting, visualising, and ensuring the reproducibility of quantitative research.

# Quantitative methods



# Quantitative methods



# Questions

Justin van Dijk

j.t.vandijk@ucl.ac.uk

